

EXERCISES

In Exercises 1–8, decide whether or not the given set, together with the indicated operations of addition and scalar multiplication, is a (real) vector space.

1. The set $\mathbb{R}^2$, with the usual addition but with scalar multiplication defined by $r[x, y] = [ry, rx]$.
2. The set $\mathbb{R}^2$, with the usual scalar multiplication but with addition defined by $[x, y] \dot{+} [r, s] = [y + s, x + r]$.
3. The set $\mathbb{R}^2$, with addition defined by $[x, y] \dot{+} [a, b] = [x + a + 1, y + b]$ and with scalar multiplication defined by $r[x, y] = [rx + r - 1, ry]$.
4. The set of all 2×2 matrices, with the usual scalar multiplication but with addition defined by $A \dot{+} B = O$, the 2×2 zero matrix.
5. The set of all 2×2 matrices, with the usual addition but with scalar multiplication defined by $rA = O$, the 2×2 zero matrix.
6. The set F of all functions mapping $\mathbb{R}$ into $\mathbb{R}$, with scalar multiplication defined as in Example 7 but with addition defined by $(f \dot{+} g)(x) = \max\{f(x), g(x)\}$.
7. The set F of all functions mapping $\mathbb{R}$ into $\mathbb{R}$, with scalar multiplication defined as in Example 7 but with addition defined by $(f \dot{+} g)(x) = f(x) + 2g(x)$.
8. The set F of all functions mapping $\mathbb{R}$ into $\mathbb{R}$, with scalar multiplication defined as in Example 7 but with addition defined by $(f \dot{+} g)(x) = 2f(x) + 2g(x)$.

In Exercises 9–16, determine whether the given set is closed under the usual operations of addition and scalar multiplication, and is a (real) vector space.

9. The set of all upper-triangular $n \times n$ matrices.

10. The set of all 2×2 matrices of the form

$$\begin{bmatrix} \times & 1 \\ 1 & \times \end{bmatrix},$$

where each $\times$ may be any scalar.

11. The set of all diagonal $n \times n$ matrices.
12. The set of all 3×3 matrices of the form

$$\begin{bmatrix} \times & 0 & \times \\ 0 & \times & 0 \\ \times & 0 & \times \end{bmatrix},$$

where each $\times$ may be any scalar.

13. The set $\{0\}$ consisting only of the number 0.
14. The set $\mathbb{Q}$ of all rational numbers.
15. The set $\mathbb{C}$ of complex numbers; that is,

$$\mathbb{C} = \{a + b\sqrt{-1} \mid a, b \text{ in } \mathbb{R}\},$$

with the usual addition of complex numbers and with scalar multiplication defined in the usual way by $r(a + b\sqrt{-1}) = ra + rb\sqrt{-1}$ for any numbers a, b , and r in $\mathbb{R}$.

16. The set P_n of all polynomials in x , with real coefficients and of degree less than or equal to n , together with the zero polynomial.
17. Your answer to Exercise 3 should be that $\mathbb{R}^2$ with the given operations is a vector space.
 - a. Describe the “zero vector” in this vector space.
 - b. Explain why the relations $r[0, 0] = [r - 1, 0] \neq [0, 0]$ do not violate property (5) of Theorem 3.1.
18. Mark each of the following True or False.
 - a. Matrix multiplication is a vector-space operation on the set $M_{m \times n}$ of all $m \times n$ matrices.
 - b. Matrix multiplication is a vector-space operation on the set M_n of all square $n \times n$ matrices.
 - c. Multiplication of any vector by the zero scalar always yields the zero vector.
 - d. Multiplication of a nonzero vector by a nonzero scalar never yields the zero vector.
 - e. No vector is its own additive inverse.

- f. The zero vector is the only vector that is its own additive inverse.
 - g. Multiplication of two scalars is of no concern in the definition of a vector space.
 - h. One of the axioms for a vector space relates addition of scalars, multiplication of a vector by scalars, and addition of vectors.
 - i. Every vector space has at least two vectors.
 - j. Every vector space has at least one vector.
19. Prove property 2 of Theorem 3.1.
 20. Prove property 3 of Theorem 3.1.
 21. Prove property 5 of Theorem 3.1.
 22. Prove property 6 of Theorem 3.1.
 23. Let V be a vector space. Prove that, if v is in V and if r is a scalar and if $rv = 0$, then either $r = 0$ or $v = 0$.
 24. Let V be a vector space and let v and w be nonzero vectors in V . Prove that if v is not a scalar multiple of w , then v is not a scalar multiple of $v + w$.
 25. Let V be a vector space, and let v and w be vectors in V . Prove that there is a *unique* vector x in V such that $x + v = w$.

Exercises 26–29 are based on the optional subsection on the universality of function spaces.

26. Using the discussion of function spaces at the end of this section, explain how we can view the Euclidean vector space $\mathbb{R}^m$ and the vector space $M_{m,n}$ of all $m \times n$ matrices as essentially the same vector space with just a different notation for the vectors.
27. Repeat Exercise 26 for the vector space $M_{2,6}$ of 2×6 matrices and the vector space $M_{3,4}$ of 3×4 matrices.
28. If you worked Exercise 16 correctly, you found that the polynomials in x of degree at most n do form a vector space P_n . Explain how P_n and $\mathbb{R}^{n+1}$ can be viewed as essentially the same vector space, with just a different notation for the vectors.
29. Referring to the three preceding exercises, list the vector spaces $\mathbb{R}^{24}$, $\mathbb{R}^{25}$, $\mathbb{R}^{26}$, P_{24} , P_{25} , P_{26} , $M_{4,7}$, $M_{3,8}$, $M_{3,9}$, $M_{2,12}$, $M_{2,13}$, $M_{4,6}$, and $M_{3,5}$ in two or more columns in such a way that any two vector spaces listed in the same column can be viewed as the same vector space with just different notation for vectors, but two vector spaces that appear in different columns cannot be so viewed.

3.2

BASIC CONCEPTS OF VECTOR SPACES

We now extend the terminology we developed in Chapters 1 and 2 for the Euclidean spaces $\mathbb{R}^n$ to general vector spaces V . That is, we discuss linear combinations of vectors, the span of vectors, subspaces, dependent and independent vectors, bases, and dimension. Definitions of most of these concepts, and the theorems concerning them, can be lifted with minor changes from Chapters 1 and 2, replacing “in $\mathbb{R}^n$ ” wherever it occurs by “in a vector space V .” Where we do make changes, the reasons for them are explained:

Linear Combinations, Spans, and Subspaces

The major change from Chapter 1 is that our vector spaces may now be so large that they cannot be spanned by a *finite* number of vectors. Each Euclidean space $\mathbb{R}^n$ and each subspace of $\mathbb{R}^n$ can be spanned by a finite set of vectors, but this is not the case for a general vector space. For example, no

finite set of polynomials can span the space P of all polynomials in x , because a finite set of polynomials cannot contain polynomials of arbitrarily high degree. Because we can add only a finite number of vectors, our definition of a linear combination in Chapter 1 will be unchanged. However, surely we want to consider the space P of all polynomials to be spanned by the monomials in the infinite set $\{1, x, x^2, x^3, \dots\}$, because every polynomial is a linear combination of these monomials. Thus we must modify the definition of the span of vectors to include the case where the number of vectors may be infinite.

DEFINITION 3.2 Linear Combinations

Given vectors $v_1, v_2, \dots, v_k$ in a vector space V and scalars $r_1, r_2, \dots, r_k$ in $\mathbb{R}$, the vector

$$r_1 v_1 + r_2 v_2 + \dots + r_k v_k$$

is a **linear combination** of the vectors $v_1, v_2, \dots, v_k$ with **scalar coefficients** $r_1, r_2, \dots, r_k$.

DEFINITION 3.3 Span of a Subset X of V

Let X be a subset of a vector space V . The **span** of X is the set of all linear combinations of vectors in X , and is denoted by $\text{sp}(X)$. If X is a finite set, so that $X = \{v_1, v_2, \dots, v_k\}$, then we also write $\text{sp}(X)$ as $\text{sp}(v_1, v_2, \dots, v_k)$. If $W = \text{sp}(X)$, the vectors in X **span** or **generate** W . If $V = \text{sp}(X)$ for some finite subset X of V , then V is **finitely generated**.

HISTORICAL NOTE A COORDINATE-FREE TREATMENT of vector-space concepts appeared in 1862 in the second version of *Ausdehnungslehre* (*The Calculus of Extension*) by Hermann Grassman (1809–1877). In this version he was able to suppress somewhat the philosophical bias that had made his earlier work so unreadable and to concentrate on his new mathematical ideas. These included the basic ideas of the theory of n -dimensional vector spaces, including linear combinations, linear independence, and the notions of a subspace and a basis. He developed the idea of the dimension of a subspace as the maximal number of linearly independent vectors and proved the fundamental relation for two subspaces V and W that $\dim(V + W) = \dim(V) + \dim(W) - \dim(V \cap W)$.

Grassmann's notions derived from the attempt to translate geometric ideas about n -dimensional space into the language of algebra without dealing with coordinates, as is done in ordinary analytic geometry. He was the first to produce a complete system in which such concepts as points, line segments, planes, and their analogues in higher dimensions are represented as single elements. Although his ideas were initially difficult to understand, ultimately they entered the mathematical mainstream in such fields as vector analysis and the exterior algebra. Grassmann himself, unfortunately, never attained his goal of becoming a German university professor, spending most of his professional life as a mathematics teacher at a gymnasium (high school) in Stettin. In the final decades of his life, he turned away from mathematics and established himself as an expert in linguistics.

In general, we suppress Euclidean space illustrations here, because they are already familiar from Chapter 1.

ILLUSTRATION 1 Let P be the vector space of all polynomials, and let $M = \{1, x, x^2, x^3, \dots\}$ be this subset of monomials. Then $P = \text{sp}(M)$. Our remarks above Definition 3.2 indicate that P is not finitely generated. ■

ILLUSTRATION 2 Let $M_{m,n}$ be the vector space of all $m \times n$ matrices, and let E be the set consisting of the matrices E_{ij} , where E_{ij} is the $m \times n$ matrix having entry 1 in the i th row and j th column and entries 0 elsewhere. There are mn of these matrices in the set E . Then $M_{m,n} = \text{sp}(E)$ and is finitely generated. ■

The notion of closure of a subset of a vector space V under vector addition or scalar multiplication is the same as for a subset of $\mathbb{R}^n$. Namely, a subset W of a vector space V is **closed under vector addition** if for all $u, v \in W$ the sum $u + v$ is in W . If for all $v \in W$ and all scalars r we have $rv \in W$, then W is **closed under scalar multiplication**.

We will call a subset W of V a **subspace** precisely when it is nonempty and closed under vector addition and scalar multiplication, just as in Definition 1.16. However, we really should define it here in a different fashion, reflecting the fact that we have given an *axiomatic definition* of a vector space. A vector space is just one of many mathematical structures that are defined axiomatically. (Some other such axiomatic structures are groups, rings, fields, topological spaces, fiber bundles, sheaves, and manifolds.) A substructure is always understood to be a subset of the original structure set that satisfies, all by itself, the axioms for that type of structure, using *inherited* features, such as operations of addition and scalar multiplication, from the original structure. We give a definition of a subspace of a vector space V that reflects this point of view, and then prove as a theorem that a subset of V is a subspace if and only if it is nonempty and closed under vector addition and scalar multiplication.

DEFINITION 3.4 Subspace

A subset W of a vector space V is a **subspace** of V if W itself fulfills the requirements of a vector space, where addition and scalar multiplication of vectors in W produce the same vectors as these operations did in V .

In order for a nonempty subset W of a vector space V to be a subspace, the subset (together with the operations of vector addition and scalar multiplication) must form a self-contained system. That is, any addition or scalar multiplication using vectors in the subset W must always yield a vector that lies again in W . Then taking any v in W , we see that $0v = 0$ and $(-1)v = -v$ are also in W . The eight properties A1–A4 and S1–S4 required of a vector space in Definition 3.1 are sure to be true for the subset, because they hold in all of V .

That is, if W is nonempty and is closed under addition and scalar multiplication, it is sure to be a vector space in its own right. We have arrived at an efficient test for determining whether a subset is a subspace of a vector space.

THEOREM 3.2 Test for a Subspace

A subset W of a vector space V is a subspace of V if and only if W is nonempty and satisfies the following two conditions:

1. If v and w are in W , then $v + w$ is in W . Closure under vector addition
2. If r is any scalar in $\mathbb{R}$ and v is in W , then rv is in W . Closure under scalar multiplication

Condition (2) of Theorem 3.2 with $r = 0$ shows that the zero vector lies in every subspace. Recall that a subspace of $\mathbb{R}^n$ always contains the origin.

The entire vector space V satisfies the conditions of Theorem 3.2. That is, V is a subspace of itself. Other subspaces of V are called **proper subspaces**. One such subspace is the subset $\{0\}$, consisting of only the zero vector. We call $\{0\}$ the **zero subspace** of V .

Note that if V is a vector space and X is any nonempty subset of V , then $\text{sp}(X)$ is a subspace of V , because the sum of two linear combinations of vectors in X is again a linear combination of vectors in X , as is any scalar multiple of such a linear combination. Thus the closure conditions of Theorem 3.2 are satisfied. A moment of thought shows that $\text{sp}(X)$ is the smallest subspace of V containing all the vectors in X .

ILLUSTRATION 3 The space P of all polynomials in x is a subspace of the vector space P_∞ of power series in x , described in Example 4 of Section 3.1. Exercise 16 in Section 3.1 shows that the set consisting of all polynomials in x of degree at most n , together with the zero polynomial, is a vector space P_n . This space P_n is a subspace both of P and of P_∞ . ■

ILLUSTRATION 4 The set of invertible $n \times n$ matrices is not a subspace of the vector space M_n of all $n \times n$ matrices, because the sum of two invertible matrices may not be invertible; also, the zero matrix is not invertible. ■

ILLUSTRATION 5 The set of all upper-triangular $n \times n$ matrices is a subspace of the space M_n of all $n \times n$ matrices, because sums and scalar multiples of upper-triangular matrices are again upper triangular. ■

ILLUSTRATION 6 Let F be the vector space of all functions mapping $\mathbb{R}$ into $\mathbb{R}$. Because sums and scalar multiples of continuous functions are continuous, the subset C of F consisting of all continuous functions mapping $\mathbb{R}$ into $\mathbb{R}$ is a subspace of F . Because sums and scalar multiples of differentiable functions are differentiable, the subset D of F consisting of all differentiable functions mapping $\mathbb{R}$ into

$\mathbb{R}$ is also a subspace of F . Because every differentiable function is continuous, we see that D is also a subspace of C . Let D_∞ be the set of all functions mapping $\mathbb{R}$ into $\mathbb{R}$ that have derivatives of all orders. Note that D_∞ is closed under addition and scalar multiplication and is a subspace of D , C , and F . ■

EXAMPLE 1 Let F be the vector space of all functions mapping $\mathbb{R}$ into $\mathbb{R}$. Show that the set S of all solutions in F of the differential equation

$$f'' + f = 0$$

is a subspace of F .

SOLUTION We note that the zero function in F is a solution, and so the set S is nonempty. If f and g are in S , then $f'' + f = 0$ and $g'' + g = 0$, and so $(f + g)'' + (f + g) = f'' + g'' + f + g = (f'' + f) + (g'' + g) = 0 + 0$, which shows that S is closed under addition. Similarly, $(rf)'' + rf = r f'' + rf = r(f'' + f) = r0 = 0$, so S is closed under scalar multiplication. Thus S is a subspace of F . ■

The preceding example is a special case of a general theorem stating that all solutions in F of a homogeneous linear differential equation form a subspace of F . We ask you to write out the proof of the general theorem in Exercise 40. Recall that all solutions of the homogeneous linear system $Ax = 0$, where A is an $m \times n$ matrix, form a subspace of $\mathbb{R}^n$.

Independence

We wish to extend the notions of dependence and independence that were given in Chapter 2. We restricted our consideration to finite sets of vectors in Chapter 2 because we can't have more than n vectors in an independent subset of $\mathbb{R}^n$. In this chapter, we have to worry about larger sets, because it may take an infinite set to span a vector space V . Recall that the vector space P of all polynomials cannot be spanned by a finite set of vectors. We make the following slight modification to Definition 2.1.

DEFINITION 3.5 Linear Dependence and Independence

Let X be a set of vectors in a vector space V . A **dependence relation** in this set X is an equation of the form

$$r_1 \mathbf{v}_1 + r_2 \mathbf{v}_2 + \cdots + r_k \mathbf{v}_k = \mathbf{0}, \quad \text{some } r_j \neq 0$$

where $\mathbf{v}_i \in X$ for $i = 1, 2, \dots, k$. If such a dependence relation exists, then X is a **linearly dependent** set of vectors. Otherwise, the set X of vectors is **linearly independent**.

ILLUSTRATION 7 The subset $\{1, x, x^2, \dots, x^n, \dots\}$ of monomials in the vector space P of all polynomials is an independent set. ■

ILLUSTRATION 8 The subset $\{\sin^2 x, \cos^2 x, 1\}$ of the vector space F of all functions mapping $\mathbb{R}$ into $\mathbb{R}$ is dependent. A dependence relation is

$$1(\sin^2 x) + 1(\cos^2 x) + (-1)1 = 0.$$

Similarly, the subset $\{\sin^2 x, \cos^2 x, \cos 2x\}$ is dependent, because we have the trigonometric identity $\cos 2x = \cos^2 x - \sin^2 x$. Thus $1(\cos 2x) + (-1)(\cos^2 x) + 1(\sin^2 x) = 0$ is a dependence relation. ■

We know a mechanical procedure for determining whether a finite set of vectors in $\mathbb{R}^n$ is independent. We simply put the vectors as column vectors in a matrix and reduce the matrix to row-echelon form. The set is independent if and only if every column in the matrix contains a pivot. There is no such mechanical procedure for determining whether a finite set of vectors in a general vector space V is independent. We illustrate two methods that are used in function spaces in the next two examples.

EXAMPLE 2 Show that $\{\sin x, \cos x\}$ is an independent set of functions in the space F of all functions mapping $\mathbb{R}$ into $\mathbb{R}$.

SOLUTION We show that there is no dependence relation of the form

$$r(\sin x) + s(\cos x) = 0, \quad (1)$$

where the 0 on the right of the equation is the function that has the value 0 for all x . If Eq. (1) holds for all x , then setting $x = 0$ and $x = \pi/2$, we obtain the linear system

$$r(0) + s(1) = 0 \quad \text{Setting } x = 0$$

$$r(1) + s(0) = 0, \quad \text{Setting } x = \frac{\pi}{2}$$

whose only solution is $r = s = 0$. Thus Eq. (1) holds only if $r = s = 0$, and so the functions $\sin x$ and $\cos x$ are independent. ■

From Example 2, we see that one way to try to show independence of k functions $f_1(x), f_2(x), \dots, f_k(x)$ is to substitute k different values of x in a dependence relation format

$$r_1 f_1(x) + r_2 f_2(x) + \dots + r_k f_k(x) = 0.$$

This will lead to a homogeneous system of k equations in the k unknowns $r_1, r_2, \dots, r_k$. If that system has only the zero solution, then the functions are independent. If there is a nontrivial solution, we can't draw any conclusion. For example, substituting $x = 0$ and $x = \pi$ in Eq. (1) in Example 2 yields the system

$$r(0) + s = 0 \quad \text{Setting } x = 0$$

$$r(0) - s = 0, \quad \text{Setting } x = \pi$$

which has a nontrivial solution—for example, $r = 10$ and $s = 0$. This occurs because we just chose the wrong values for x . The values 0 and $\pi/2$ for x do demonstrate independence, as we saw in Example 2.

EXAMPLE 3 Show that the functions e^x and e^{2x} are independent in the vector space F of all functions mapping $\mathbb{R}$ into $\mathbb{R}$.

SOLUTION We set up the dependence relation format

$$re^x + se^{2x} = 0,$$

and try to determine if we must have $r = s = 0$. Illustrating a different technique than in Example 2, we write this equation and its derivative:

$$re^x + se^{2x} = 0$$

$$re^x + 2se^{2x} = 0. \quad \text{Differentiating}$$

Setting $x = 0$ in both equations, we obtain the homogeneous linear system

$$r + s = 0$$

$$r + 2s = 0,$$

which has only the trivial solution $r = s = 0$. Thus the functions are independent. ■

In summary, we can try to show independence of functions by starting with an equation in dependence relation format, and then substituting different values of the variable, or differentiating (possibly several times) and substituting values, or a combination of both, to obtain a square homogeneous system with the coefficients in the dependence relation format as unknowns. If the system has only the zero solution, the functions are independent. If there are nontrivial solutions, we can't come to a conclusion without more work.

Bases and Dimension

Recall that we defined a subset $\{w_1, w_2, \dots, w_k\}$ to be a *basis* for the subspace $W = \text{sp}(w_1, w_2, \dots, w_k)$ of $\mathbb{R}^n$ if every vector in W can be expressed *uniquely* as a linear combination of $w_1, w_2, \dots, w_k$. Theorem 1.15 shows that to demonstrate this uniqueness, we need only show that the only linear combination that yields the zero vector is the one with all coefficients 0—that is, the uniqueness condition can be replaced by the condition that the set $\{w_1, w_2, \dots, w_k\}$ be independent. This led us to an alternate characterization of a basis for W (Theorem 2.1) as a subset of W that is *independent* and that *spans* W . It is this alternate description that is traditional for general vector spaces. The uniqueness condition then becomes a theorem; it remains the most important aspect of a basis and forms the foundation for the next section of our text. For a general vector space, we may need an *infinite* set of vectors to form a basis: for example, a basis for the space P of all polynomials is the set

$\{1, x, x^2, \dots, x^n, \dots\}$ of monomials. The following definition takes this possibility into account. Also, because a subspace of a vector space is again a vector space, it is unnecessary now to explicitly include the word "subspace" in the definition.

DEFINITION 3.6 Basis for a Vector Space

Let V be a vector space. A set of vectors in V is a **basis** for V if the following conditions are met:

1. The set of vectors spans V .
2. The set of vectors is linearly independent.

ILLUSTRATION 9 The set $X = \{1, x, x^2, \dots, x^n, \dots\}$ of monomials is a basis for the vector space P of all polynomials. It is not a basis for the vector space P_∞ of formal power series in x , discussed in Example 4 of Section 3.1, because a series $\sum_{n=0}^{\infty} a_n x^n$ cannot be expressed as a *finite* sum of scalar multiples of the monomials unless all but a finite number of the coefficients a_n are 0. For example, $1 + x + x^2 + x^3 + \dots$ is not a finite sum of monomials. Remember that all linear combinations are *finite* sums.

The vector space P_n of polynomials of degree at most n , together with the zero polynomial, has as a basis $\{1, x, x^2, \dots, x^n\}$. ■

We now prove as a theorem the uniqueness which was the defining criterion in Definition 1.17 in Section 1.6. Namely, we show that a subset B of nonzero vectors in a vector space V is a basis for V if and only if each vector v in V can be expressed *uniquely* in the form

$$v = r_1 b_1 + r_2 b_2 + \dots + r_k b_k \quad (2)$$

for scalars r_i and vectors b_i in B . Because B can be an infinite set, we need to elaborate on the meaning of uniqueness. Suppose that there are two expressions in the form of Eq. (2) for v . The two expressions might involve some of the same vectors from B and might involve some different vectors from B . The important thing is that each involves only a finite number of vectors. Thus if we take all vectors in B appearing in one expression or the other, or in both, we have just a finite list of vectors to be concerned with. We may assume that both expressions contain each vector in this list by inserting any missing vector with a zero coefficient. Assuming now that Eq. (2) is the result of this adjustment of the first expression for v , the second expression for v may be written as

$$v = s_1 b_1 + s_2 b_2 + \dots + s_k b_k. \quad (3)$$

Uniqueness asserts that $s_i = r_i$ for each i .

THEOREM 3.3 Unique Combination Criterion for a Basis

Let B be a set of nonzero vectors in a vector space V . Then B is a basis for V if and only if each vector v in V can be *uniquely* expressed in the form of Eq. (2) for scalars r_i and vectors $b_i \in B$.

PROOF Assume that B is a basis for V . Condition 1 of Definition 3.6 tells us that a vector v in V can be expressed in the form of Eq. (2). Suppose now that v can also be written in the form of Eq. (3). Subtracting Eq. (3) from Eq. (2), we obtain

$$(r_1 - s_1)b_1 + (r_2 - s_2)b_2 + \cdots + (r_k - s_k)b_k = 0.$$

Because B is independent, we see that $r_1 - s_1 = 0$, $r_2 - s_2 = 0$, $\dots$, $r_k - s_k = 0$, and so $r_i = s_i$ for each i . Thus we have established uniqueness.

Now assume that each vector in V can be expressed uniquely in the form of Eq. (2). In particular, this is true of the zero vector. This means that no dependence relation in B is possible. That is, if

$$r_1b_1 + r_2b_2 + \cdots + r_kb_k = 0$$

for vectors b_i in B and scalars r_i , then, because we always have

$$0b_1 + 0b_2 + \cdots + 0b_k = 0,$$

each $r_i = 0$ by uniqueness. This shows that B is independent. Because B also generates V (by hypothesis), it is a basis. $\blacktriangle$

Dimension

We will have to restrict our treatment here to **finitely generated vector spaces**—that is, those that can be spanned by a finite number of vectors. We can argue just as we did for $\mathbb{R}^n$ that every finitely generated vector space V contains a basis.* Namely, if $V = \text{sp}(v_1, v_2, \dots, v_k)$, then we can examine the vectors v_i in turn, and delete at each step any that can be expressed as linear combinations of those that remain. We would also like to know that any two bases have the same number of elements, so that we can have a well-defined concept of dimension for a finitely generated vector space. The main tool is Theorem 2.2 on page 130, which we can restate for general vector spaces, rather than for subspaces of $\mathbb{R}^n$.

*If we are willing to assume the **Axiom of Choice**:

Given a collection of nonempty sets, no two of which have an element in common, there exists a "choice set" C that contains exactly one element from each set in the collection.

then we can prove that every vector space has a basis, and that given a vector space V , every basis has the same number of elements, although we may be totally unable to actually specify a basis. This kind of work is regarded as magnificent mathematics by some and as abstract nonsense by others. For example, using the Axiom of Choice, we can prove that the space P_∞ of all formal power series in x has a basis. But we are still unable to exhibit one! It has been shown that the Axiom of Choice is independent of the other axioms of set theory.

THEOREM 3.4 Relative Size of Spanning and Independent Sets

Let V be a vector space. Let $w_1, w_2, \dots, w_k$ be vectors in V that span V , and let $v_1, v_2, \dots, v_m$ be vectors in V that are independent. Then $k \geq m$.

PROOF The proof is the same, word for word, as the proof of Theorem 2.2. $\blacktriangle$

It is not surprising that the proof of the preceding theorem is the same as that of Theorem 2.2. The next section will show that we can expect Chapter 2 arguments to be valid whenever we deal just with finitely generated vector spaces.

The same arguments as those in the corollary to Theorem 2.2 give us the following corollary to Theorem 3.4.

COROLLARY Invariance of Dimension for Finitely Generated Spaces

Let V be a finitely generated vector space. Then any two bases of V have the same number of elements.

We can now rewrite the definition of dimension for finitely generated vector spaces.

DEFINITION 3.7 Dimension of a Finitely Generated Vector Space

Let V be a finitely generated vector space. The number of elements in a basis for V is the dimension of V , and is denoted by $\dim(V)$.

ILLUSTRATION 10 Let P_n be the vector space of polynomials in x of degree at most n . Because $\{1, x, x^2, \dots, x^n\}$ is a basis for P_n , we see that $\dim(P_n) = n + 1$. $\blacksquare$

ILLUSTRATION 11 The set E of matrices E_{ij} in Illustration 2 is a basis for the vector space $M_{m,n}$ of all $m \times n$ matrices, so $\dim(M_{m,n}) = mn$. $\blacksquare$

By the same arguments that we used for $\mathbb{R}^n$ (page 132), we see that for a finitely generated vector space V , every independent set of vectors in V can be enlarged, if necessary, to a basis. Also, if $\dim(V) = k$, then every independent set of k vectors in V is a basis for V , and every set of k vectors that span V is a basis for V . (See Theorem 2.3 on page 133.)

EXAMPLE 4 Determine whether $S = \{1 - x, 2 - 3x^2, x + 2x^2\}$ is a basis for the vector space P_2 of polynomials of degree at most 2, together with the zero polynomial.

SOLUTION We know that $\dim(P_2) = 3$ because $\{1, x, x^2\}$ is a basis for P_2 . Thus S will be a basis if and only if S is an independent set. We can rewrite the dependence relation format

$$r(1 - x) + s(2 - 3x^2) + t(x + 2x^2) = 0$$

as

$$(r + 2s)1 + (-r + t)x + (-3s + 2t)x^2 = 0.$$

Because $\{1, x, x^2\}$ is independent, this relation can hold if and only if

$$\begin{aligned} r + 2s &= 0 \\ -r + t &= 0 \\ -3s + 2t &= 0. \end{aligned}$$

Reducing the coefficient matrix of this homogeneous system, we obtain

$$\begin{bmatrix} 1 & 2 & 0 \\ -1 & 0 & 1 \\ 0 & -3 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 1 \\ 0 & -3 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & \frac{7}{2} \end{bmatrix}.$$

We see at once that the homogeneous system with this coefficient matrix has only the trivial solution. Thus no dependence relation exists, and so S is an independent set with the necessary number of vectors, and is thus a basis. ■

EXAMPLE 5 Find a basis for the vector space P_3 (polynomials of degree at most 3, and 0) containing the polynomials $x^2 + 1$ and $x^2 - 1$.

SOLUTION First we observe that the two given polynomials are independent because neither is a scalar multiple of the other. The vectors

$$x^2 + 1, x^2 - 1, 1, x, x^2, x^3$$

generate P_3 because the last four of these form a basis for P_3 . We can reduce this list of vectors to a basis by deleting any vector that is a linear combination of others in the list, being sure to retain the first two. For this example, it is actually easier to notice that surely x is not in $\text{sp}(x^2 + 1, x^2 - 1)$ and x^3 is not in $\text{sp}(x^2 - 1, x^2 + 1, x)$. Thus the set $\{x^2 + 1, x^2 - 1, x, x^3\}$ is independent. Because $\dim(P_3) = 4$ and this independent set contains four vectors, it must be a basis for P_3 . Alternatively, we could have deleted 1 and x by noticing that

$$1 = \frac{1}{2}(x^2 + 1) - \frac{1}{2}(x^2 - 1) \quad \text{and} \quad x^2 = \frac{1}{2}(x^2 + 1) + \frac{1}{2}(x^2 - 1). \quad \blacksquare$$

SUMMARY

1. A subset W of a vector space V is a *subspace* of V if and only if it is nonempty and satisfies the two closure properties:
 $v + w$ is contained in W for all vectors v and w in W , and
 rv is contained in W for all vectors v in W and all scalars r .
2. Let X be a subset of a vector space V . The set $\text{sp}(X)$ of all linear combinations of vectors in X is a subspace of V called the *span* of X , or the *subspace of V generated by the vectors in X* . It is the smallest subspace of V containing all the vectors in X .
3. A vector space V is *finitely generated* if $V = \text{sp}(X)$ for some set $X = \{v_1, v_2, \dots, v_k\}$ containing only a finite number of vectors in V .
4. A set X of vectors in a vector space V is *linearly dependent* if there exists a dependence relation

$$r_1 v_1 + r_2 v_2 + \dots + r_k v_k = 0, \quad \text{at least one } r_j \neq 0,$$

where each $v_i \in X$ and each $r_i \in \mathbb{R}$. The set X is *linearly independent* if no such dependence relation exists, and so a linear combination of vectors in X is the zero vector only if all scalar coefficients are zero.

5. A set B of vectors in a vector space V is a *basis* for V if B spans V and is independent.
6. A subset B of nonzero vectors in a vector space V is a basis for V if and only if every nonzero vector in V can be expressed as a linear combination of vectors in B in a *unique* way.
7. If X is a finite set of vectors spanning a vector space V , then X can be reduced, if necessary, to a basis for V by deleting in turn any vector that can be expressed as a linear combination of those remaining.
8. If a vector space V has a finite basis, then all bases for V have the same number of vectors. The number of vectors in a basis for V is the *dimension* of V , denoted by $\dim(V)$.
9. The following are equivalent for n vectors in a vector space V where $\dim(V) = n$.
 - a. The vectors are linearly independent.
 - b. The vectors generate V .

EXERCISES

In Exercises 1–6, determine whether the indicated subset is a subspace of the given vector space.

1. The set of all polynomials of degree greater than 3 together with the zero polynomial in

the vector space P of all polynomials with coefficients in $\mathbb{R}$

2. The set of all polynomials of degree 4 together with the zero polynomial in the vector space P of all polynomials in x

3. The set of all functions f such that $f(0) = 1$ in the vector space F of all functions mapping $\mathbb{R}$ into $\mathbb{R}$
4. The set of all functions f such that $f(1) = 0$ in the vector space F of all functions mapping $\mathbb{R}$ into $\mathbb{R}$
5. The set of all functions f in the vector space W of differentiable functions mapping $\mathbb{R}$ into $\mathbb{R}$ (see Illustration 6) such that $f'(2) = 0$
6. The set of all functions f in the vector space W of differentiable functions mapping $\mathbb{R}$ into $\mathbb{R}$ (see Illustration 6) such that f has derivatives of all orders
7. Let F be the vector space of functions mapping $\mathbb{R}$ into $\mathbb{R}$. Show that
 - a. $\text{sp}(\sin^2 x, \cos^2 x)$ contains all constant functions,
 - b. $\text{sp}(\sin^2 x, \cos^2 x)$ contains the function $\cos 2x$,
 - c. $\text{sp}(7, \sin^2 2x)$ contains the function $8 \cos 4x$.
8. Let P be the vector space of polynomials. Prove that $\text{sp}(1, x) = \text{sp}(1 + 2x, x)$. [HINT: Show that each of these subspaces is a subset of the other.]
9. Let V be a vector space, and let v_1 and v_2 be vectors in V . Follow the hint of Exercise 8 to prove that
 - a. $\text{sp}(v_1, v_2) = \text{sp}(v_1, 2v_1 + v_2)$,
 - b. $\text{sp}(v_1, v_2) = \text{sp}(v_1 + v_2, v_1 - v_2)$.
10. Let $v_1, v_2, \dots, v_k$ and $w_1, w_2, \dots, w_m$ be vectors in a vector space V . Give a necessary and sufficient condition, involving linear combinations, for

$$\text{sp}(v_1, v_2, \dots, v_k) = \text{sp}(w_1, w_2, \dots, w_m).$$

In Exercises 11–13, determine whether the given set of vectors is dependent or independent.

11. $\{x^2 - 1, x^2 + 1, 4x, 2x - 3\}$ in P
12. $\{1, 4x + 3, 3x - 4, x^2 + 2, x - x^2\}$ in P
13. $\{1, \sin^2 x, \cos 2x, \cos^2 x\}$ in F

In Exercises 14–19, use the technique discussed following Example 3 to determine whether the given set of functions in the vector space F is independent or dependent.

14. $\{\sin x, \cos x\}$
15. $\{1, x, x^2\}$
16. $\{\sin x, \sin 2x, \sin 3x\}$
17. $\{\sin x, \sin(-x)\}$
18. $\{e^{2x}, e^{3x}, e^{4x}\}$
19. $\{1, e^x + e^{-x}, e^x - e^{-x}\}$

In Exercises 20 and 21, determine whether or not the given set of vectors is a basis for the indicated vector space.

20. $\{x, x^2 + 1, (x - 1)^2\}$ for P_2
21. $\{x, (x + 1)^2, (x - 1)^2\}$ for P_2

In Exercises 22–24, find a basis for the given subspace of the vector space.

22. $\text{sp}(x^2 - 1, x^2 + 1, 4, 2x - 3)$ in P
23. $\text{sp}(1, 4x + 3, 3x - 4, x^2 + 2, x - x^2)$ in P
24. $\text{sp}(1, \sin^2 x, \cos 2x, \cos^2 x)$ in F
25. Mark each of the following True or False.
 - a. The set consisting of the zero vector is a subspace for every vector space.
 - b. Every vector space has at least two distinct subspaces.
 - c. Every vector space with a nonzero vector has at least two distinct subspaces.
 - d. If $\{v_1, v_2, \dots, v_n\}$ is a subset of a vector space V , then v_i is in $\text{sp}(v_1, v_2, \dots, v_n)$ for $i = 1, 2, \dots, n$.
 - e. If $\{v_1, v_2, \dots, v_n\}$ is a subset of a vector space V , then the sum $v_i + v_j$ is in $\text{sp}(v_1, v_2, \dots, v_n)$ for all choices of i and j from 1 to n .
 - f. If $u + v$ lies in a subspace W of a vector space V , then both u and v lie in W .
 - g. Two subspaces of a vector space V may have empty intersection.
 - h. If S is independent, each vector in V can be expressed uniquely as a linear combination of vectors in S .
 - i. If S is independent and generates V , each vector in V can be expressed uniquely as a linear combination of vectors in S .
 - j. If each vector in V can be expressed uniquely as a linear combination of vectors in S , then S is an independent set.

26. Let V be a vector space. Mark each of the following True or False.
- a. Every independent set of vectors in V is a basis for the subspace the vectors span.
 - b. If $\{v_1, v_2, \dots, v_n\}$ generates V , then each $v \in V$ is a linear combination of the vectors in this set.
 - c. If $\{v_1, v_2, \dots, v_n\}$ generates V , then each $v \in V$ is a unique linear combination of the vectors in this set.
 - d. If $\{v_1, v_2, \dots, v_n\}$ generates V and is independent, then each $v \in V$ is a unique linear combination of the vectors in this set.
 - e. If $\{v_1, v_2, \dots, v_n\}$ generates V , then this set of vectors is independent.
 - f. If each vector in V is a unique linear combination of the vectors in the set $\{v_1, v_2, \dots, v_n\}$, then this set is independent.
 - g. If each vector in V is a unique linear combination of the vectors in the set $\{v_1, v_2, \dots, v_n\}$, then this set is a basis for V .
 - h. All vector spaces having a basis are finitely generated.
 - i. Every independent subset of a finitely generated vector space V is a part of some basis for V .
 - j. Any two bases in a finite-dimensional vector space V have the same number of elements.
27. Let W_1 and W_2 be subspaces of a vector space V . Prove that the intersection $W_1 \cap W_2$ is also a subspace of V .
28. Let $W_1 = \text{sp}((1, 2, 3), (2, 1, 1))$ and $W_2 = \text{sp}((1, 0, 1), (3, 0, -1))$ in $\mathbb{R}^3$. Find a set of generating vectors for $W_1 \cap W_2$.
29. Let V be a vector space with basis $\{v_1, v_2, v_3\}$. Prove that $\{v_1, v_1 + v_2, v_1 + v_2 + v_3\}$ is also a basis for V .
30. Let V be a vector space with basis $\{v_1, v_2, \dots, v_n\}$, and let $W = \text{sp}(v_3, v_4, \dots, v_n)$. If $w = r_1 v_1 + r_2 v_2$ is in W , show that $w = 0$.
31. Let $\{v_1, v_2, v_3\}$ be a basis for a vector space V . Prove that the vectors $w_1 = v_1 + v_2$, $w_2 = v_2 + v_3$, $w_3 = v_1 - v_3$ do not generate V .
32. Let $\{v_1, v_2, v_3\}$ be a basis for a vector space V . Prove that, if w is not in $\text{sp}(v_1, v_2)$, then $\{v_1, v_2, w\}$ is also a basis for V .
33. Let $\{v_1, v_2, \dots, v_n\}$ be a basis for a vector space V , and let $w = t_1 v_1 + t_2 v_2 + \dots + t_k v_k$, with $t_k \neq 0$. Prove that
- $$\{v_1, v_2, \dots, v_{k-1}, w, v_{k+1}, \dots, v_n\}$$
- is a basis for V .
34. Let W and U be subspaces of a vector space V , and let $W \cap U = \{0\}$. Let $\{w_1, w_2, \dots, w_k\}$ be a basis for W , and let $\{u_1, u_2, \dots, u_m\}$ be a basis for U . Prove that, if each vector v in V is expressible in the form $v = w + u$ for $w \in W$ and $u \in U$, then
- $$\{w_1, w_2, \dots, w_k, u_1, u_2, \dots, u_m\}$$
- is a basis for V .
35. Illustrate Exercise 34 with nontrivial subspaces W and U of $\mathbb{R}^5$.
36. Prove that, if W is a subspace of an n -dimensional vector space V and $\dim(W) = n$, then $W = V$.
37. Let $v_1, v_2, \dots, v_n$ be a list of nonzero vectors in a vector space V such that no vector in this list is a linear combination of its predecessors. Show that the vectors in the list form an independent set.
38. Exercise 37 indicates that a finite generating set for a vector space can be reduced to a basis by deleting, from left to right in a list of the vectors, each vector that is a linear combination of its predecessors. Use this technique to find a basis for the subspace
- $$\text{sp}(x^2 + 1, x^2 + x - 1, 3x - 6, x^3 + x^2 + 1, x^3)$$
- of the polynomial space P .
39. We once watched a speaker in a lecture derive the equation $f(x) \sin x + g(x) \cos x = 0$, and then say, "Now everyone knows that $\sin x$ and $\cos x$ are independent functions, so $f(x) = 0$ and $g(x) = 0$." Was the statement correct or incorrect? Give a proof or a counterexample.

40. A homogeneous linear n th-order differential equation has the form

$$f_n(x)y^{(n)} + f_{n-1}(x)y^{(n-1)} + \cdots + f_2(x)y'' + f_1(x)y' + f_0(x)y = 0.$$

Show that the set of all solutions of this equation that lie in the space F of all functions mapping $\mathbb{R}$ into $\mathbb{R}$ is a subspace of F .

41. Referring to Exercise 40, suppose that the differential equation

$$f_n(x)y^{(n)} + f_{n-1}(x)y^{(n-1)} + \cdots + f_2(x)y'' + f_1(x)y' + f_0(x)y = g(x)$$

does have a solution $y = p(x)$ in the space F of all functions mapping $\mathbb{R}$ into $\mathbb{R}$. By analogy with Theorem 1.18 on p. 97, describe the structure of the set of solutions of this equation that lie in F .

42. Solve the differential equation $y' = 2x$ and describe your solution in terms of your answer to Exercise 41, or in terms of the answer in the back of the text.

It is a theorem of differential equations that if the functions $f_i(x)$ of the differential equation in Exercise 40 are all constant, then all the solutions of the equation lie in the vector space F of all functions mapping $\mathbb{R}$ into $\mathbb{R}$ and form a subspace of F of dimension n . Thus every solution can be written as a linear combination of n independent functions in F that form a basis for the solution space.

In Exercises 43–45, use your knowledge of calculus and the solution of Exercise 41 to describe the solution set of the given differential equation. You should be able to work these problems without having had a course in differential equations, using the hints.

43. a. $y'' + y = 0$ [HINT: You need to find two independent functions such that when you differentiate twice, you get the negative of the function you started with.]
 b. $y'' + y = x$ [HINT: Find one solution by experimentation.]
44. a. $y'' - 4y = 0$ [HINT: What two independent functions, when differentiated twice, give 4 times the original function?]
 b. $y'' - 4y = x$ [HINT: Find one solution by experimentation.]
45. a. $y^{(3)} - 9y' = 0$ [HINT: Try to find values of m such that $y = e^{mx}$ is a solution.]
 b. $y^{(3)} - 9y' = x^2 + 2x$ [HINT: Find one solution by experimentation.]
46. Let S be any set and let F be the set of all functions mapping S into $\mathbb{R}$. Let W be the subset of F consisting of all functions $f \in F$ such that $f(s) = 0$ for all but a finite number of elements s in S .
 a. Show that W is a subspace of F .
 b. What condition must be satisfied to have $W = F$?
47. Referring to Exercise 46, describe a basis B for the subspace W of F . Explain why B is not a basis for F unless $F = W$.

3.3

COORDINATIZATION OF VECTORS

Much of the work in this text is phrased in terms of the Euclidean vector spaces $\mathbb{R}^n$ for $n = 1, 2, 3, \dots$. In this section we show that, for finite-dimensional vector spaces, no loss of generality results from restricting ourselves to the spaces $\mathbb{R}^n$. Specifically, we will see that if a vector space V has dimension n , then V can be coordinatized so that it will look just like $\mathbb{R}^n$. We can then work with these coordinates by utilizing the matrix techniques we have developed for the space $\mathbb{R}^n$. Throughout this section, we consider V to be a finite-dimensional vector space.

Ordered Bases

The vector $[2, 5]$ in $\mathbb{R}^2$ can be expressed in terms of the standard basis vectors as $2\mathbf{e}_1 + 5\mathbf{e}_2$. The components of $[2, 5]$ are precisely the coefficients of these basis vectors. The vector $[2, 5]$ is different from the vector $[5, 2]$, just as the point $(2, 5)$ is different from the point $(5, 2)$. We regard the standard basis vectors as having a natural order, $\mathbf{e}_1 = [1, 0]$ and $\mathbf{e}_2 = [0, 1]$. In a nonzero vector space V with a basis $B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$, there is usually no natural order for the basis vectors. For example, the vectors $\mathbf{b}_1 = [-1, 5]$ and $\mathbf{b}_2 = [3, 2]$ form a basis for $\mathbb{R}^2$, but there is no natural order for these vectors. If we want the vectors to have an order, we must specify their order. By convention, set notation does not denote order; for example, $\{\mathbf{b}_1, \mathbf{b}_2\} = \{\mathbf{b}_2, \mathbf{b}_1\}$. To describe order, we use parentheses, $(\)$, in place of set braces, $\{\ }$; we are used to paying attention to order in the notation $(\mathbf{b}_1, \mathbf{b}_2)$. We denote an *ordered basis* of n vectors in V by $B = (\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n)$. For example, the standard basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ of $\mathbb{R}^3$ gives rise to six different ordered bases—namely,

$$(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3) \quad (\mathbf{e}_2, \mathbf{e}_1, \mathbf{e}_3) \quad (\mathbf{e}_3, \mathbf{e}_1, \mathbf{e}_2) \quad (\mathbf{e}_1, \mathbf{e}_3, \mathbf{e}_2) \quad (\mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_1) \quad (\mathbf{e}_3, \mathbf{e}_2, \mathbf{e}_1).$$

These correspond to the six possible orders for the unit coordinate vectors. The ordered basis $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$ is the *standard ordered basis* for $\mathbb{R}^3$, and in general, the basis $E = (\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n)$ is the *standard ordered basis* for $\mathbb{R}^n$.

Coordinatization of Vectors

Let V be a finite-dimensional vector space, and let $B = (\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n)$ be a basis for V . By Theorem 3.3, every vector $\mathbf{v}$ in V can be expressed in the form

$$\mathbf{v} = r_1\mathbf{b}_1 + r_2\mathbf{b}_2 + \cdots + r_n\mathbf{b}_n$$

for unique scalars $r_1, r_2, \dots, r_n$. We associate the vector $[r_1, r_2, \dots, r_n]$ in $\mathbb{R}^n$ with $\mathbf{v}$. This gives us a way of coordinatizing V .

DEFINITION 3.8 Coordinate Vector Relative to an Ordered Basis

Let $B = (\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n)$ be an ordered basis for a finite-dimensional vector space V , and let

$$\mathbf{v} = r_1\mathbf{b}_1 + r_2\mathbf{b}_2 + \cdots + r_n\mathbf{b}_n.$$

The vector $[r_1, r_2, \dots, r_n]$ is the *coordinate vector* of $\mathbf{v}$ relative to the ordered basis B , and is denoted by $\mathbf{v}_B$.

ILLUSTRATION 1 Let P_n be the vector space of polynomials of degree at most n . There are two natural choices for an ordered basis for P_n —namely,

$$B = (1, x, x^2, \dots, x^n) \quad \text{and} \quad B' = (x^n, x^{n-1}, \dots, x^2, x, 1).$$

Taking $n = 4$, we see that for the polynomial $p(x) = -x + x^3 + 2x^4$ we have

$$p(x)_B = [0, -1, 0, 1, 2] \quad \text{and} \quad p(x)_{B'} = [2, 1, 0, -1, 0]. \quad \blacksquare$$

EXAMPLE 1 Find the coordinate vectors of $[1, -1]$ and of $[-1, -8]$ relative to the ordered basis $B = ([1, -1], [1, 2])$ of $\mathbb{R}^2$.

SOLUTION We see that $[1, -1]_B = [1, 0]$, because

$$[1, -1] = 1[1, -1] + 0[1, 2].$$

To find $[-1, -8]_B$, we must find r_1 and r_2 such that $[-1, -8] = r_1[1, -1] + r_2[1, 2]$. Equating components of this vector equation, we obtain the linear system

$$r_1 + r_2 = -1$$

$$-r_1 + 2r_2 = -8.$$

The solution of this system is $r_1 = 2$, $r_2 = -3$, so we have $[-1, -8]_B = [2, -3]$. Figure 3.1 indicates the geometric meaning of these coordinates. ■

EXAMPLE 2 Find the coordinate vector of $[1, 2, -2]$ relative to the ordered basis $B = ([1, 1, 1], [1, 2, 0], [1, 0, 1])$ in $\mathbb{R}^3$.

SOLUTION We must express $[1, 2, -2]$ as a linear combination of the basis vectors in B . Working with column vectors, we must solve the equation

$$r_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + r_2 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + r_3 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}$$

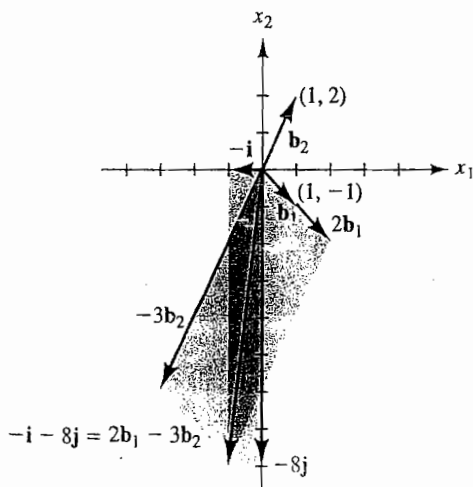


FIGURE 3.1

$$[-1, -8]_B = [2, -3].$$

for r_1 , r_2 , and r_3 . We find the unique solution by a Gauss–Jordan reduction:

$$\begin{aligned} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 2 & 0 & 2 \\ 1 & 0 & 1 & -2 \end{array} \right] &\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 1 \\ 0 & -1 & 0 & -3 \end{array} \right] \\ &\sim \left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & -1 & -2 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & -4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 2 \end{array} \right]. \end{aligned}$$

Therefore, $[1, 2, -2]_B = [-4, 3, 2]$. ■

We now box the procedure illustrated by Example 2.

Finding the Coordinate Vector of $\mathbf{v}$ in $\mathbb{R}^n$ Relative to an Ordered Basis $B = (\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n)$

Step 1: Writing vectors as column vectors, form the augmented matrix $[\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n, \mathbf{v}]$.

Step 2: Use a Gauss–Jordan reduction to obtain the augmented matrix $[I, \mathbf{v}_B]$, where I is the $n \times n$ identity matrix and $\mathbf{v}_B$ is the desired coordinate vector.

Coordinatization of a Finite-Dimensional Vector Space

We can coordinatize a finite-dimensional vector space V by selecting an ordered basis $B = (\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n)$ and associating with each vector in V its unique coordinate vector relative to B . This gives a one-to-one correspondence between all the vectors in V and all the vectors in $\mathbb{R}^n$. To show that we may now work in $\mathbb{R}^n$ rather than in V , we have to show that the vector-space operations of vector addition and scalar multiplication in V are mirrored by those operations on coordinate vectors in $\mathbb{R}^n$. That is, we must show that

$$(\mathbf{v} + \mathbf{w})_B = \mathbf{v}_B + \mathbf{w}_B \quad \text{and} \quad (t\mathbf{v})_B = t\mathbf{v}_B \quad (1)$$

for all vectors $\mathbf{v}$ and $\mathbf{w}$ in V and for all scalars t in $\mathbb{R}$. To do this, suppose that

$$\mathbf{v} = r_1\mathbf{b}_1 + r_2\mathbf{b}_2 + \cdots + r_n\mathbf{b}_n$$

and

$$\mathbf{w} = s_1\mathbf{b}_1 + s_2\mathbf{b}_2 + \cdots + s_n\mathbf{b}_n.$$

Because

$$\mathbf{v} + \mathbf{w} = (r_1 + s_1)\mathbf{b}_1 + (r_2 + s_2)\mathbf{b}_2 + \cdots + (r_n + s_n)\mathbf{b}_n,$$

we see that the coordinate vector of $\mathbf{v} + \mathbf{w}$ is

$$\begin{aligned}(\mathbf{v} + \mathbf{w})_B &= [r_1 + s_1, r_2 + s_2, \dots, r_n + s_n] \\&= [r_1, r_2, \dots, r_n] + [s_1, s_2, \dots, s_n] \\&= \mathbf{v}_B + \mathbf{w}_B,\end{aligned}$$

which is the sum of the coordinate vectors of $\mathbf{v}$ and of $\mathbf{w}$. Similarly, for any scalar t , we have

$$\begin{aligned}t\mathbf{v} &= t(r_1\mathbf{b}_1 + r_2\mathbf{b}_2 + \dots + r_n\mathbf{b}_n) \\&= (tr_1)\mathbf{b}_1 + (tr_2)\mathbf{b}_2 + \dots + (tr_n)\mathbf{b}_n,\end{aligned}$$

so the coordinate vector of $t\mathbf{v}$ is

$$\begin{aligned}(t\mathbf{v})_B &= [tr_1, tr_2, \dots, tr_n] \\&= t[r_1, r_2, \dots, r_n] = t\mathbf{v}_B.\end{aligned}$$

This completes the demonstration of relations (1). These relations tell us that, when we rename the vectors in V by coordinates relative to B , the resulting vector space of coordinates—namely, $\mathbb{R}^n$ —has the same vector-space structure as V . Whenever the vectors in a vector space V can be renamed to make V appear structurally identical to a vector space W , we say that V and W are *isomorphic vector spaces*. Our discussion shows that every real vector space having a basis of n vectors is isomorphic to $\mathbb{R}^n$. For example, the space P_n of all polynomials of degree at most n is isomorphic to $\mathbb{R}^{n+1}$, because P_n has an ordered basis $B = (x^n, x^{n-1}, \dots, x^2, x, 1)$ of $n+1$ vectors. Each polynomial

$$a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

can be renamed by its coordinate vector

$$[a_n, a_{n-1}, \dots, a_1, a_0]$$

relative to B . The adjective *isomorphic* is used throughout algebra to signify that two algebraic structures are identical except in the names of their elements.

For a vector space V isomorphic to a vector space W , all of the algebraic properties of vectors in V that can be derived solely from the axioms of a vector space correspond to identical properties in W . However, we cannot expect other features—such as whether the vectors are functions, matrices, or n -tuples—to be carried over from one space to the other. But a generating set of vectors or an independent set of vectors in one space corresponds to a set with the same property in the other space. Here is an example showing how we can simplify computations in a finite-dimensional vector space V , by working instead in $\mathbb{R}^n$.

EXAMPLE 3 Determine whether $x^2 - 3x + 2$, $3x^2 + 5x - 4$, and $7x^2 + 21x - 16$ are independent in the vector space P_2 of polynomials of degree at most 2.

SOLUTION We take $B = (x^2, x, 1)$ as an ordered basis for P_2 . The coordinate vectors relative to B of the given polynomials are

$$\begin{aligned}(x^2 - 3x + 2)_B &= [1, -3, 2], \\(3x^2 + 5x - 4)_B &= [3, 5, -4], \\(7x^2 + 21x - 16)_B &= [7, 21, -16].\end{aligned}$$

We can determine whether the polynomials are independent by determining whether the corresponding coordinate vectors in $\mathbb{R}^3$ are independent. We set up the usual matrix, with these vectors as column vectors, and then we row-reduce it, obtaining

$$\begin{bmatrix} 1 & 3 & 7 \\ -3 & 5 & 21 \\ 2 & -4 & -16 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 7 \\ 0 & 14 & 42 \\ 0 & -10 & -30 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 7 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}.$$

Because the third column in the echelon form has no pivot, these three coordinate vectors in $\mathbb{R}^3$ are dependent, and so the three polynomials are dependent. ■

Continuing Example 3, to further illustrate working with coordinates in $\mathbb{R}^n$, we can reduce the final matrix further to

$$\begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}.$$

If we imagine a partition between the second and third columns, we see that

$$[7, 21, -16] = -2[1, -3, 2] + 3[3, 5, -4].$$

Thus

$$7x^2 + 21x - 16 = -2(x^2 - 3x + 2) + 3(3x^2 + 5x - 4).$$

EXAMPLE 4 It can be shown that the set $\{1, \sin x, \sin 2x, \dots, \sin nx\}$ is an independent subset of the vector space F of all functions mapping $\mathbb{R}$ into $\mathbb{R}$. Find a basis for the subspace of F spanned by the functions

$$\begin{aligned}f_1(x) &= 3 - \sin x + 3 \sin 2x - \sin 3x + 5 \sin 4x, \\f_2(x) &= 1 + 2 \sin x + 4 \sin 2x - \sin 4x \\f_3(x) &= -1 + 5 \sin x + 5 \sin 2x + \sin 3x - 7 \sin 4x \\f_4(x) &= 3 \sin 2x - \sin 4x.\end{aligned}$$

SOLUTION We see that all these functions lie in the subspace W of F given by $W = \text{sp}(1, \sin x, \sin 2x, \sin 3x, \sin 4x)$, and we take

$$B = (1, \sin x, \sin 2x, \sin 3x, \sin 4x)$$

as an ordered basis for this subspace. Working with coordinates relative to B , the problem reduces to finding a basis for the subspace of $\mathbb{R}^5$ spanned by $[3, -1, 3, -1, 5]$, $[1, 2, 4, 0, -1]$, $[-1, 5, 5, 1, -7]$, and $[0, 0, 3, 0, -1]$. We reduce the matrix having these as column vectors, and begin this by switching minus the fourth row with the first row to create the pivot 1 in the first column. We obtain

$$\begin{bmatrix} 3 & 1 & -1 & 0 \\ -1 & 2 & 5 & 0 \\ 3 & 4 & 5 & 3 \\ -1 & 0 & 1 & 0 \\ 5 & -1 & -7 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 2 & 4 & 0 \\ 0 & 4 & 8 & 3 \\ 0 & 1 & 2 & 0 \\ 0 & -1 & -2 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Because the first, second, and fourth columns have pivots, we should keep the first, second, and fourth of the original column vectors, so that the set $\{f_1(x), f_2(x), f_4(x)\}$ is a basis for the subspace W . ■

We give an example showing the utility of different bases in the study of polynomial functions. We know that a polynomial function

$$y = p(x) = a_n x^n + \cdots + a_2 x^2 + a_1 x + a_0$$

has a graph that passes through the origin if $a_0 = 0$. If both a_0 and a_1 are zero, then the graph not only goes through the origin, but is quite flat there; indeed, if $a_2 \neq 0$, then the function behaves approximately like $a_2 x^2$ very near $x = 0$, because for very small values of x , such as 0.00001, the value of x^2 is much greater than the values of x^3, x^4 , and other higher powers of x . If $a_2 = 0$ also, but $a_3 \neq 0$, then the function behaves very much like $a_3 x^3$ for such values of x very close to 0, etc. If instead of studying a polynomial function near $x = 0$, we want to study it near some other x -value, say near $x = a$, then we would like to express the polynomial as a linear combination of powers $(x - a)^i$ —that is,

$$p(x) = b_n(x - a)^n + \cdots + b_2(x - a)^2 + b_1(x - a) + b_0.$$

Both $B = (x^n, \dots, x^2, x, 1)$ and $B' = ((x - a)^n, \dots, (x - a)^2, x - a, 1)$ are ordered bases for the space P_n of polynomials of degree at most n . (We leave the demonstration that B' is a basis as Exercise 20.) We give an example illustrating a method for expressing the polynomial $x^3 + x^2 - x - 1$ as a linear combination of $(x + 1)^3, (x + 1)^2, x + 1$, and 1.

EXAMPLE 5 Find the coordinate vector of $p(x) = x^3 + x^2 - x - 1$ relative to the ordered basis $B' = ((x + 1)^3, (x + 1)^2, x + 1, 1)$.

SOLUTION Multiplying out the powers of $x + 1$, to express the vectors in B' in terms of our usual ordered basis $B = (x^3, x^2, x, 1)$ for P_3 , we see that

$$B' = (x^3 + 3x^2 + 3x + 1, x^2 + 2x + 1, x + 1, 1).$$

Using coordinates relative to the ordered basis B , our problem reduces to expressing the vector $[1, 1, -1, -1]$ as a linear combination of the vectors

$[1, 3, 3, 1]$, $[0, 1, 2, 1]$, $[0, 0, 1, 1]$, and $[0, 0, 0, 1]$. Reducing the matrix corresponding to the associated linear system, we obtain

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 3 & 1 & 0 & 0 & 1 \\ 3 & 2 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & -1 \end{array} \right] \sim \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & -2 \\ 0 & 2 & 1 & 0 & -4 \\ 0 & 1 & 1 & 1 & -2 \end{array} \right] \sim \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & -2 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{array} \right] \sim \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & -2 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array} \right]$$

Thus the required coordinate vector is $p(x)_B = [1, -2, 0, 0]$, and so

$$x^3 + x^2 - x - 1 = (x + 1)^3 - 2(x + 1)^2. \quad \blacksquare$$

Linear algebra is not the only tool that can be used to solve the problem in Example 5. Exercise 13 suggests a polynomial algebra solution, and Exercise 16 describes a calculus solution.

SUMMARY

Let V be a vector space with basis $\{b_1, b_2, \dots, b_n\}$.

1. $B = (b_1, b_2, \dots, b_n)$ is an ordered basis; the vectors are regarded as being in a specified order in this n -tuple notation.
2. Each vector v in V has a unique expression as a linear combination:

$$v = r_1 b_1 + r_2 b_2 + \cdots + r_n b_n.$$

3. The vector $v_B = [r_1, r_2, \dots, r_n]$ for the uniquely determined scalars r_i in the preceding equation (summary item 2) is the coordinate vector of v relative to B .
4. The vector space V can be coordinatized, using summary item 3, so that V is isomorphic to $\mathbb{R}^n$.

EXERCISES

In Exercises 1–10, find the coordinate vector of the given vector relative to the indicated ordered basis.

1. $[-1, 1]$ in $\mathbb{R}^2$ relative to $\{[0, 1], [1, 0]\}$
2. $[-2, 4]$ in $\mathbb{R}^2$ relative to $\{[0, -2], [-\frac{1}{2}, 0]\}$
3. $[4, 6, 2]$ in $\mathbb{R}^3$ relative to $\{[2, 0, 0], [0, 1, 1], [0, 0, 1]\}$
4. $[4, -2, 1]$ in $\mathbb{R}^3$ relative to $\{[0, 1, 1], [2, 0, 0], [0, 3, 0]\}$
5. $[3, 13, -1]$ in $\mathbb{R}^3$ relative to $\{[1, 3, -2], [4, 1, 3], [-1, 2, 0]\}$
6. $[9, 6, 11, 0]$ in $\mathbb{R}^4$ relative to $\{[1, 0, 1, 0], [2, 1, 1, -1], [0, 1, 1, -1], [2, 1, 3, 1]\}$

7. $x^3 + x^2 - 2x + 4$ in P_3 relative to $(1, x^2, x, x^3)$
8. $x^3 + 3x^2 - 4x + 2$ in P_3 relative to $(x, x^2 - 1, x^3, 2x^2)$
9. $x + x^4$ in P_4 relative to $(1, 2x - 1, x^3 + x^4, 2x^3, x^2 + 2)$

10. $\begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$ in M_2 relative to

$$\left\{ \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 0 & 3 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} \right\}$$

11. Find the coordinate vector of the polynomial $x^3 - 4x^2 + 3x + 7$ relative to the ordered basis $B' = ((x - 2)^3, (x - 2)^2, (x - 2), 1)$ of

the vector space P_3 of polynomials of degree at most 3. Use the method illustrated in Example 5.

12. Find the coordinate vector of the polynomial $4x^3 - 9x^2 + x$ relative to the ordered basis $B' = ((x-1)^3, (x-1)^2, (x-1), 1)$ of the vector space P_3 of polynomials of degree at most 3. Use the method illustrated in Example 5.
13. Example 5 showed how to use *linear algebra* to rewrite the polynomial $p(x) = x^3 + x^2 - x - 1$ in powers of $x+1$ rather than in powers of x . This exercise indicates a *polynomial algebra* solution to this problem. Replace x in $p(x)$ by $[(x+1) - 1]$, and expand using the binomial theorem, keeping the $(x+1)$ intact. Check your answer with that in Example 5.
14. Repeat Exercise 11 using the polynomial algebra method indicated in Exercise 13.
15. Repeat Exercise 12 using the polynomial algebra method indicated in Exercise 13.
16. Example 5 showed how to use *linear algebra* to rewrite the polynomial $p(x) = x^3 + x^2 - x - 1$ in powers of $x+1$ rather than in powers of x . This exercise indicates a *calculus* solution to this problem. Form the equation

$$x^3 + x^2 - x - 1 = b_3(x+1)^3 + b_2(x+1)^2 + b_1(x+1) + b_0.$$
 Find b_0 by substituting $x = -1$ in this equation. Then equate the derivatives of both sides, and substitute $x = -1$ to find b_1 . Continue differentiating both sides and substituting $x = -1$ to find b_2 and b_3 . Check your answer with that in Example 5.
17. Repeat Exercise 11 using the calculus method indicated in Exercise 16.
18. Repeat Exercise 12 using the calculus method indicated in Exercise 16.
19. a. Prove that $\{1, \sin x, \cos x, \sin 2x, \cos 2x\}$ is an independent set of functions in the vector space F of all functions mapping $\mathbb{R}$ into $\mathbb{R}$.
b. Find a basis for the subspace of F generated by the functions

$$f_1(x) = 1 - 2 \sin x + 4 \cos x - \sin 2x - 3 \cos 2x,$$

$$f_2(x) = 2 - 3 \sin x - \cos x + 4 \sin 2x + 5 \cos 2x$$

$$f_3(x) = 5 - 8 \sin x + 2 \cos x + 7 \sin 2x + 7 \cos 2x$$

$$f_4(x) = -1 + 14 \cos x - 11 \sin 2x - 19 \cos 2x$$

20. Prove that for every positive integer n and every $a \in \mathbb{R}$, the set

$$\{(x-a)^n, (x-a)^{n-1}, \dots, (x-a)^2, x-a, 1\}$$
 is a basis for the vector space P_n of polynomials of degree at most n .
21. Find the polynomial in P_2 whose coordinate vector relative to the ordered basis $B = (x+x^2, x-x^2, 1+x)$ is $[3, 1, 2]$.
22. Let V be a nonzero finite-dimensional vector space. Mark each of the following True or False.
 - a. The vector space V is isomorphic to $\mathbb{R}^n$ for some positive integer n .
 - b. There is a unique coordinate vector associated with each vector $v \in V$.
 - c. There is a unique coordinate vector associated with each vector $v \in V$ relative to a basis for V .
 - d. There is a unique coordinate vector associated with each vector $v \in V$ relative to an ordered basis for V .
 - e. Distinct vectors in V have distinct coordinate vectors relative to the same ordered basis B for V .
 - f. The same vector in V cannot have the same coordinate vector relative to different ordered bases for V .
 - g. There are six possible ordered bases for $\mathbb{R}^3$.
 - h. There are six possible ordered bases for $\mathbb{R}^3$, consisting of the standard unit coordinate vectors in $\mathbb{R}^3$.
 - i. A reordering of elements in an ordered basis for V corresponds to a similar reordering of components in coordinate vectors with respect to the basis.
 - j. Addition and multiplication by scalars in V can be computed in terms of coordinate vectors with respect to any fixed ordered basis for V .

3.4

LINEAR TRANSFORMATIONS

Linear transformations mapping $\mathbb{R}^n$ into $\mathbb{R}^m$ were defined in Section 2.3. Now that we have considered vectors in more general spaces than $\mathbb{R}^n$, it is natural to extend the notion to linear transformations of other vector spaces, not necessarily finite-dimensional. In this section, we introduce linear transformations in a general setting. Recall that a linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a function that satisfies

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}) \quad (1)$$

and

$$T(r\mathbf{u}) = rT(\mathbf{u}) \quad (2)$$

for all vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ and for all scalars r .

Linear Transformations $T: V \rightarrow V'$

The definition of a linear transformation of a vector space V into a vector space V' is practically identical to the definition for the Euclidean vector spaces in Section 2.3. We need only replace $\mathbb{R}^n$ by V and $\mathbb{R}^m$ by V' .

DEFINITION 3.9 Linear Transformation

A function T that maps a vector space V into a vector space V' is a **linear transformation** if it satisfies two criteria:

1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$, **Preservation of addition**
2. $T(r\mathbf{u}) = rT(\mathbf{u})$, **Preservation of scalar multiplication**

for all vectors $\mathbf{u}$ and $\mathbf{v}$ in V and for all scalars r in $\mathbb{R}$.

Exercise 35 shows that the two conditions of Definition 3.9 may be combined into the single condition

$$T(r\mathbf{u} + s\mathbf{v}) = rT(\mathbf{u}) + sT(\mathbf{v}) \quad (3)$$

for all vectors $\mathbf{u}$ and $\mathbf{v}$ in V and for all scalars r and s in $\mathbb{R}$. Mathematical induction can be used to verify the analogous relation for n summands:

$$T(r_1\mathbf{v}_1 + r_2\mathbf{v}_2 + \cdots + r_n\mathbf{v}_n) = r_1T(\mathbf{v}_1) + r_2T(\mathbf{v}_2) + \cdots + r_nT(\mathbf{v}_n). \quad (4)$$

We remind you of some terminology and notation defined in Section 2.3 for functions in general and linear transformations in particular. We state things here in the language of linear transformations. For a linear transformation $T: V \rightarrow V'$, the set V is the **domain** of T and the set V' is the **codomain** of T . If W is a subset of V , then $T[W] = \{T(\mathbf{w}) \mid \mathbf{w} \in W\}$ is the **image** of W under T . In particular, $T[V]$ is the **range** of T . Similarly, if W' is a subset of V' , then

$T^{-1}[W'] = \{v \in V \mid T(v) \in W'\}$ is the **inverse image** of W' under T . In particular, $T^{-1}[\{0'\}]$ is the **kernel** of T , denoted by $\ker(T)$. It consists of all of the vectors in V that T maps into $0'$.

Let V, V' , and V'' be vector spaces, and let $T: V \rightarrow V'$ and $T': V' \rightarrow V''$ be linear transformations. The **composite transformation** $T' \circ T: V \rightarrow V''$ is defined by $(T' \circ T)(v) = T'(T(v))$ for v in V . Exercise 36 shows that $T' \circ T$ is again a linear transformation.

EXAMPLE 1 Let F be the vector space of all functions $f: \mathbb{R} \rightarrow \mathbb{R}$, and let D be its subspace of all differentiable functions. Show that differentiation is a linear transformation of D into F .

SOLUTION Let $T: D \rightarrow F$ be defined by $T(f) = f'$, the derivative of f . Using the familiar rules

$$(f + g)' = f' + g' \quad \text{and} \quad (rf)' = r(f')$$

for differentiation from calculus, we see that

$$T(f + g) = (f + g)' = f' + g' = T(f) + T(g)$$

and

$$T(rf) = (rf)' = r(f') = rT(f)$$

for all functions f and g in D and scalars r . In other words, these two rules for differentiating a sum of functions and for differentiating a scalar times a function constitute precisely the assertion that differentiation is a linear transformation. ■

EXAMPLE 2 Let F be the vector space of all functions $f: \mathbb{R} \rightarrow \mathbb{R}$, and let c be in $\mathbb{R}$. Show that the evaluation function $T: F \rightarrow \mathbb{R}$ defined by $T(f) = f(c)$, which maps each function f in F into its value at c , is a linear transformation.

HISTORICAL NOTE THE CONCEPT OF A LINEAR SUBSTITUTION dates back to the eighteenth century. But it was only after physicists became used to dealing with vectors that the idea of a function of vectors became explicit. One of the founders of vector analysis, Oliver Heaviside (1850–1925), introduced the idea of a linear vector operator in one of his works on electromagnetism in 1885. He defined it using coordinates: $\vec{B}$ comes from $\vec{H}$ by a linear vector operator if, when $\vec{B}$ has components B_1, B_2, B_3 and $\vec{H}$ has components H_1, H_2, H_3 , there are numbers μ_{ij} for $i, j = 1, 2, 3$, where

$$B_1 = \mu_{11}H_1 + \mu_{12}H_2 + \mu_{13}H_3$$

$$B_2 = \mu_{21}H_1 + \mu_{22}H_2 + \mu_{23}H_3$$

$$B_3 = \mu_{31}H_1 + \mu_{32}H_2 + \mu_{33}H_3.$$

In his lectures at Yale, which were published in 1901, J. Willard Gibbs called this same transformation a *linear vector function*. But he also defined this more abstractly as a continuous function f such that $f(\vec{v} + \vec{w}) = f(\vec{v}) + f(\vec{w})$. A fully abstract definition, exactly like Definition 3.9, was given by Hermann Weyl in *Space-Time-Matter* (1918).

Oliver Heaviside was a self-taught expert on mathematical physics who played an important role in the development of electromagnetic theory and especially its practical applications. In 1901 he predicted the existence of a reflecting ionized region surrounding the earth; the existence of this layer, now called the *ionosphere*, was soon confirmed.

SOLUTION We show that T preserves addition and scalar multiplication. If f and g are functions in the vector space F , then, evaluating $f + g$ at c , we obtain

$$(f + g)(c) = f(c) + g(c).$$

Therefore,

$$\begin{aligned} T(f + g) &= (f + g)(c) && \text{Definition of } T \\ &= f(c) + g(c) && \text{Definition of } f + g \text{ in } F \\ &= T(f) + T(g). && \text{Definition of } T \end{aligned}$$

This shows that T preserves addition. In a similar manner, the computation

$$\begin{aligned} T(rf) &= (rf)(c) && \text{Definition of } T \\ &= r(f(c)) && \text{Definition of } rf \text{ in } F \\ &= r(T(f)) && \text{Definition of } T \end{aligned}$$

shows that T preserves scalar multiplication. ■

EXAMPLE 3 Let $C_{a,b}$ be the vector space of all continuous functions mapping the closed interval $a \leq x \leq b$ of $\mathbb{R}$ into $\mathbb{R}$. Show that $T: C_{a,b} \rightarrow \mathbb{R}$ defined by $T(f) = \int_a^b f(x) dx$ is a linear transformation.

SOLUTION From properties of the definite integral, we know that for $f, g \in C_{a,b}$ and for any scalar r , we have

$$T(f + g) = \int_a^b [f(x) + g(x)] dx = \int_a^b f(x) dx + \int_a^b g(x) dx = T(f) + T(g)$$

and

$$T(rf) = \int_a^b rf(x) dx = r \int_a^b f(x) dx = rT(f).$$

This shows that T is indeed a linear transformation. ■

EXAMPLE 4 Let C be the vector space of all continuous functions mapping $\mathbb{R}$ into $\mathbb{R}$. Let $a \in \mathbb{R}$ and let $T_a: C \rightarrow C$ be defined by $T_a(f) = \int_a^x f(t) dt$. Show that T_a is a linear transformation.

SOLUTION This follows from the same properties of the integral that we used in the solution of the preceding exercise. From calculus, we know that the range of T_a is actually a subset of the vector space of differentiable functions mapping $\mathbb{R}$ into $\mathbb{R}$. (Theorem 3.7, which follows shortly, will show that the range is actually a subspace.) ■

EXAMPLE 5 Let D_{∞} be the space of functions mapping $\mathbb{R}$ into $\mathbb{R}$ that have derivatives of all orders, and let $a_0, a_1, a_2, \dots, a_n \in \mathbb{R}$. Show that $T: D_{\infty} \rightarrow D_{\infty}$ defined by $T(f) = a_n f^{(n)}(x) + \dots + a_2 f''(x) + a_1 f'(x) + a_0 f(x)$ is a linear transformation.

SOLUTION This follows from the fact that the i th derivative of a sum of functions is the sum of their i th derivatives—that is, $(f+g)^{(i)}(x) = f^{(i)}(x) + g^{(i)}(x)$ —and that the i th derivative of $rf(x)$ is $r f^{(i)}(x)$, together with the fact that $T(f)$ is defined to be a *linear combination* of these derivatives. (We consider $f(x)$ to be the 0th derivative.) ■

Note that the computation of $T(f)$ in Example 5 amounts to the computation of the left-hand side of the general linear differential equation with constant coefficients

$$a_n y^{(n)} + \dots + a_2 y'' + a_1 y' + a_0 y = g(x) \quad (5)$$

for $y = f(x)$. Thus, solving this differential equation amounts to finding all $f \in D_{\infty}$ such that $T(f) = g(x)$.

Properties of Linear Transformations

The two properties of linear transformations in our next theorem are useful and easy to prove.

THEOREM 3.5 Preservation of Zero and Subtraction

Let V and V' be vector spaces, and let $T: V \rightarrow V'$ be a linear transformation. Then

1. $T(\mathbf{0}) = \mathbf{0}'$, and Preservation of zero
2. $T(\mathbf{v}_1 - \mathbf{v}_2) = T(\mathbf{v}_1) - T(\mathbf{v}_2)$ Preservation of subtraction

for any vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ in V .

PROOF We establish preservation of zero by taking $r = 0$ and $\mathbf{v} = \mathbf{0}$ in condition 2 of Definition 3.9 for a linear transformation. Condition 2 and the property $0\mathbf{v} = \mathbf{0}$ (see Theorem 3.1) yield

$$T(\mathbf{0}) = T(0\mathbf{0}) = 0T(\mathbf{0}) = \mathbf{0}'.$$

Preservation of subtraction follows from Eq. (3), as follows:

$$\begin{aligned} T(\mathbf{v}_1 - \mathbf{v}_2) &= T(\mathbf{v}_1 + (-1)\mathbf{v}_2) \\ &= T(\mathbf{v}_1) + (-1)T(\mathbf{v}_2) \\ &= T(\mathbf{v}_1) - T(\mathbf{v}_2). \end{aligned}$$

EXAMPLE 6 Let D be the vector space of all differentiable functions and F the vector space of all functions mapping $\mathbb{R}$ into $\mathbb{R}$. Determine whether $T: D \rightarrow F$ defined by $T(f) = 2f'''(x) + 3f'(x) + x^2$ is a linear transformation.

SOLUTION Because for the zero constant function we have $T(0) = 2(0'') + 3(0') + x^2 = 2(0) + 3(0) + x^2 = x^2$, and x^2 is not the zero constant function, we see that T does not preserve zero, and so it is not a linear transformation. ■

Just as for a linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ discussed in Section 2.3, a linear transformation of a vector space is determined by its action on any basis for the domain. (See Exercises 40 and 41 as well.) Because our bases here may be infinite, we restate the result and give the short proof.

THEOREM 3.6 Bases and Linear Transformations

Let $T: V \rightarrow V'$ be a linear transformation, and let B be a basis for V . For any vector v in V , the vector $T(v)$ is uniquely determined by the vectors $T(b)$ for all $b \in B$. In other words, if two linear transformations have the same value at each basis vector $b \in B$, the two transformations have the same value at each vector in V ; that is, they are the same transformation.

PROOF Let T and $\bar{T}$ be two linear transformations such that $T(b_i) = \bar{T}(b_i)$ for each vector b_i in B . Let $v \in V$. Then there exist vectors $b_1, b_2, \dots, b_k$ in B and scalars $r_1, r_2, \dots, r_k$ such that

$$v = r_1 b_1 + r_2 b_2 + \cdots + r_k b_k.$$

We then have

$$\begin{aligned} T(v) &= T(r_1 b_1 + r_2 b_2 + \cdots + r_k b_k) \\ &= r_1 T(b_1) + r_2 T(b_2) + \cdots + r_k T(b_k) \\ &= r_1 \bar{T}(b_1) + r_2 \bar{T}(b_2) + \cdots + r_k \bar{T}(b_k) \\ &= \bar{T}(r_1 b_1 + r_2 b_2 + \cdots + r_k b_k) \\ &= \bar{T}(v). \end{aligned}$$

Thus T and $\bar{T}$ are the same transformation. ▲

The next theorem also generalizes results that appear in the text and exercises of Section 2.3 for linear transformations mapping $\mathbb{R}^n$ into $\mathbb{R}^m$.

THEOREM 3.7 Preservation of Subspaces

Let V and V' be vector spaces, and let $T: V \rightarrow V'$ be a linear transformation.

1. If W is a subspace of V , then $T[W]$ is a subspace of V' .
2. If W' is a subspace of V' , then $T^{-1}[W']$ is a subspace of V .

PROOF

1. Because $T(0) = 0'$, we need only show that $T[W]$ is closed under vector addition and under scalar multiplication. Let $T(\mathbf{w}_1)$ and $T(\mathbf{w}_2)$ be any vectors in $T(W)$, where $\mathbf{w}_1$ and $\mathbf{w}_2$ are vectors in W . Then

$$T(\mathbf{w}_1) + T(\mathbf{w}_2) = T(\mathbf{w}_1 + \mathbf{w}_2),$$

by preservation of addition. Now $\mathbf{w}_1 + \mathbf{w}_2$ is in W because W is itself closed under addition, and so $T(\mathbf{w}_1 + \mathbf{w}_2)$ is in $T[W]$. This shows that $T[W]$ is closed under vector addition. If r is any scalar, then $r\mathbf{w}_1$ is in W , and

$$rT(\mathbf{w}_1) = T(r\mathbf{w}_1).$$

This shows that $rT(\mathbf{w}_1)$ is in $T[W]$, and so $T[W]$ is closed under scalar multiplication. Thus, $T[W]$ is a subspace of V' .

2. Notice that $0 \in T^{-1}[W']$. Let $\mathbf{v}_1$ and $\mathbf{v}_2$ be any vectors in $T^{-1}[W']$, so that $T(\mathbf{v}_1)$ and $T(\mathbf{v}_2)$ are in W' . Then

$$T(\mathbf{v}_1 + \mathbf{v}_2) = T(\mathbf{v}_1) + T(\mathbf{v}_2)$$

is also in the subspace W' , and so $\mathbf{v}_1 + \mathbf{v}_2$ is in $T^{-1}[W']$. For any scalar r , we know that

$$rT(\mathbf{v}_1) = T(r\mathbf{v}_1),$$

and $rT(\mathbf{v}_1)$ is in W' . Thus, $r\mathbf{v}_1$ is also in $T^{-1}[W']$. This shows that $T^{-1}[W']$ is closed under addition and under scalar multiplication, and so $T^{-1}[W']$ is a subspace of V . $\blacktriangle$

The Equation $T(\mathbf{x}) = \mathbf{b}$

Let $T: V \rightarrow V'$ be a linear transformation. From Theorem 3.7, we know that $\ker(T) = T^{-1}[\{0'\}]$ is a subspace of V . This subspace is the solution set of the *homogeneous transformation equation* $T(\mathbf{x}) = 0'$. The structure of the solution set of $T(\mathbf{x}) = \mathbf{b}$ exactly parallels the structure of the solution set of $A\mathbf{x} = \mathbf{b}$ described on p. 97. Namely,

Solution Set of $T(\mathbf{x}) = \mathbf{b}$

Let $T: V \rightarrow V'$ be a linear transformation and let $T(\mathbf{p}) = \mathbf{b}$ for a particular vector $\mathbf{p}$ in V . The solution set of $T(\mathbf{x}) = \mathbf{b}$ is the set $\{\mathbf{p} + \mathbf{h} \mid \mathbf{h} \in \ker(T)\}$.

The proof is essentially the same as that of Theorem 1.18; we ask you to write it out for this case in Exercise 46. This boxed result shows that if $\ker(T) = \{0\}$, then $T(\mathbf{x}) = \mathbf{b}$ has at most one solution, and so T is **one-to-one**, meaning that $T(\mathbf{v}_1) = T(\mathbf{v}_2)$ implies that $\mathbf{v}_1 = \mathbf{v}_2$. Conversely, if T is one-to-one, then $T(\mathbf{x}) = 0'$ has only one solution—namely, 0 —so $\ker(T) = \{0\}$. We box this fact.

Condition for T to Be One-to-One

A linear transformation T is one-to-one if and only if $\ker(T) = \{0\}$.

ILLUSTRATION 1 The differential equation $y'' - 4y = x^2$ is linear with constant coefficients and is of the form of Eq. (5). In Example 5, we showed that the left-hand side of such an equation defines a linear transformation of D_∞ into itself. In differential equations, it is shown that the kernel of the transformation $T(f) = f'' - 4f$ is two-dimensional. We can check that the independent functions $h_1(x) = e^{2x}$ and $h_2(x) = e^{-2x}$ both satisfy the differential equation $y'' - 4y = 0$. Thus $\{e^{2x}, e^{-2x}\}$ is a basis for the kernel of T . All solutions of the homogeneous system are of the form $c_1 e^{2x} + c_2 e^{-2x}$. We find a particular solution $p(x)$ of $y'' - 4y = x^2$ by inspection: if $y = -x^2/4$, then the term $-4y$ yields x^2 but the second derivative of $-x^2/4$ is $-1/2$; we can kill off this unwanted $-1/2$ by taking

$$p(x) = -\frac{1}{4}x^2 - \frac{1}{8}$$

(note that the second derivative of $-1/8$ is 0). Thus the general solution of this differential equation is

$$y = c_1 e^{2x} + c_2 e^{-2x} - \frac{1}{4}x^2 - \frac{1}{8}. \quad \blacksquare$$

Invertible Transformations

In Section 2.3, we saw that, if A is an invertible $n \times n$ matrix, the linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ defined by $T(\mathbf{x}) = A\mathbf{x}$ has an inverse $T^{-1}: \mathbb{R}^n \rightarrow \mathbb{R}^n$, where $T^{-1}(\mathbf{y}) = A^{-1}\mathbf{y}$, and that both composite transformations $T^{-1} \circ T$ and $T \circ T^{-1}$ are the identity transformation of $\mathbb{R}^n$. We now extend this idea to linear transformations of vector spaces in general.

DEFINITION 3.10 Invertible Transformation

Let V and V' be vector spaces. A linear transformation $T: V \rightarrow V'$ is **invertible** if there exists a linear transformation $T^{-1}: V' \rightarrow V$ such that $T^{-1} \circ T$ is the identity transformation on V and $T \circ T^{-1}$ is the identity transformation on V' .

EXAMPLE 7 Determine whether the evaluation transformation $T: F \rightarrow \mathbb{R}$ of Example 2, where $T(f) = f(c)$ for some fixed c in $\mathbb{R}$, is invertible.

SOLUTION Consider the polynomial functions f and g , where $f(x) = 2x + c$ and $g(x) = 4x - c$. Then $T(f) = T(g) = 3c$. If T were invertible, there would have to be a linear transformation $T^{-1}: \mathbb{R} \rightarrow F$ such that $T^{-1}(3c)$ is both f and g . But this is impossible. $\blacksquare$

Example 7 illustrates that, if $T: V \rightarrow V'$ is an *invertible* linear transformation, T must satisfy the following property:

$$\text{if } \mathbf{v}_1 \neq \mathbf{v}_2, \text{ then } T(\mathbf{v}_1) \neq T(\mathbf{v}_2). \quad \text{One-to-one} \quad (6)$$

As in the argument of Example 7, if $T(\mathbf{v}_1) = T(\mathbf{v}_2) = \mathbf{v}'$ and T is invertible, there would have to be a linear transformation $T^{-1}: V' \rightarrow V$ such that $T^{-1}(\mathbf{v}')$ is simultaneously $\mathbf{v}_1$ and $\mathbf{v}_2$, which is impossible when $\mathbf{v}_1 \neq \mathbf{v}_2$.

An invertible linear transformation $T: V \rightarrow V'$ must also satisfy another property:

$$\text{if } \mathbf{v}' \text{ is in } V', \text{ then } T(\mathbf{v}) = \mathbf{v}' \text{ for some } \mathbf{v} \text{ in } V. \quad \text{Onto} \quad (7)$$

This follows at once from the fact that, for T^{-1} with the properties in Definition 3.10 and for $\mathbf{v}'$ in V' , we have $T^{-1}(\mathbf{v}') = \mathbf{v}$ for some $\mathbf{v}$ in V . But then $\mathbf{v}' = (T \circ T^{-1})(\mathbf{v}') = T(T^{-1}(\mathbf{v}')) = T(\mathbf{v})$. A transformation $T: V \rightarrow V'$ satisfying property (7) is *onto* V' ; in this case the range of T is all of V' . We have thus proved half of the following theorem.*

THEOREM 3.8 Invertibility of T

A linear transformation $T: V \rightarrow V'$ is invertible if and only if it is one-to-one and onto V' .

PROOF We have just shown that, if T is invertible, it must be one-to-one and onto V' .

Suppose now that T is one-to-one and onto V' . Because T is onto V' , for each $\mathbf{v}'$ in V' we can find $\mathbf{v}$ in V such that $T(\mathbf{v}) = \mathbf{v}'$. Because T is one-to-one, this vector $\mathbf{v}$ in V is *unique*. Let $T^{-1}: V' \rightarrow V$ be defined by $T^{-1}(\mathbf{v}') = \mathbf{v}$, where $\mathbf{v}$ is the unique vector in V such that $T(\mathbf{v}) = \mathbf{v}'$. Then

$$(T \circ T^{-1})(\mathbf{v}') = T(T^{-1}(\mathbf{v}')) = T(\mathbf{v}) = \mathbf{v}'$$

and

$$(T^{-1} \circ T)(\mathbf{v}) = T^{-1}(T(\mathbf{v})) = T^{-1}(\mathbf{v}') = \mathbf{v},$$

which shows that $T \circ T^{-1}$ is the identity map of V' and that $T^{-1} \circ T$ is the identity map of V . It only remains to show that T^{-1} is indeed a linear transformation—that is, that

$$T^{-1}(\mathbf{v}'_1 + \mathbf{v}'_2) = T^{-1}(\mathbf{v}'_1) + T^{-1}(\mathbf{v}'_2) \quad \text{and} \quad T^{-1}(r\mathbf{v}'_1) = rT^{-1}(\mathbf{v}'_1)$$

*This theorem is valid for functions in general.

for all $\mathbf{v}'_1$ and $\mathbf{v}'_2$ in V' and for all scalars r . Let $\mathbf{v}_1$ and $\mathbf{v}_2$ be the unique vectors in V such that $T(\mathbf{v}_1) = \mathbf{v}'_1$ and $T(\mathbf{v}_2) = \mathbf{v}'_2$. Remembering that $T^{-1} \circ T$ is the identity map, we have

$$\begin{aligned} T^{-1}(\mathbf{v}'_1 + \mathbf{v}'_2) &= T^{-1}(T(\mathbf{v}_1) + T(\mathbf{v}_2)) = T^{-1}(T(\mathbf{v}_1 + \mathbf{v}_2)) \\ &= (T^{-1} \circ T)(\mathbf{v}_1 + \mathbf{v}_2) = \mathbf{v}_1 + \mathbf{v}_2 = T^{-1}(\mathbf{v}'_1) + T^{-1}(\mathbf{v}'_2). \end{aligned}$$

Similarly,

$$T^{-1}(r\mathbf{v}'_1) = T^{-1}(rT(\mathbf{v}_1)) = T^{-1}(T(r\mathbf{v}_1)) = r\mathbf{v}_1 = rT^{-1}(\mathbf{v}'_1). \quad \blacktriangle$$

The proof of Theorem 3.8 shows that, if $T: V \rightarrow V'$ is invertible, the linear transformation $T^{-1}: V' \rightarrow V$ described in Definition 3.10 is unique. This transformation T^{-1} is the **inverse transformation** of T .

ILLUSTRATION 2 Let D be the vector space of differentiable functions and F the vector space of all functions mapping $\mathbb{R}$ into $\mathbb{R}$. Then $T: D \rightarrow F$, where $T(f) = f'$, the derivative of f , is not an invertible transformation. Specifically, we have $T(x) = T(x + 17) = 1$, showing that T is not one-to-one. Notice also that the kernel of T is not $\{0\}$, but consists of all constant functions. ■

ILLUSTRATION 3 Let P be the vector space of all polynomials in x and let W be the subspace of all polynomials in x with constant term 0, so that $q(0) = 0$ for $q(x) \in W$. Let $T: P \rightarrow W$ be defined by $T(p(x)) = xp(x)$. Then T is a linear transformation. Because $x p_1(x) = x p_2(x)$ if and only if $p_1(x) = p_2(x)$, we see that T is one-to-one. Every polynomial $q(x)$ in W contains no constant term, and so it can be factored as $q(x) = xp(x)$; because $T(p(x)) = q(x)$, we see that T maps P onto W . Thus T is an invertible linear transformation. ■

Isomorphism

An **isomorphism** is a linear transformation $T: V \rightarrow V'$ that is one-to-one and onto V' . Theorem 3.8 shows that isomorphisms are precisely the invertible linear transformations $T: V \rightarrow V'$. If an isomorphism T exists, then it is invertible and its inverse is also an isomorphism. The vector spaces V and V' are said to be **isomorphic** in this case. We view isomorphic vector spaces V and V' as being structurally identical in the following sense. Let $T: V \rightarrow V'$ be an isomorphism. Rename each $\mathbf{v}$ in V by the $\mathbf{v}' = T(\mathbf{v})$ in V' . Because T is one-to-one, no two different elements of V get the same name from V' , and because T is onto V' , all names in V' are used. The renamed V and V' then appear identical as sets. But they also have the same algebraic structure as vector spaces, as Figure 3.2 illustrates. We discussed a special case of this concept before Example 3 in Section 3.3, indicating informally that every finite-dimensional vector space is structurally the same as $\mathbb{R}^n$ for some n . We are now in a position to state this formally.

THEOREM 3.9 Coordinatization of Finite-Dimensional Spaces

Let V be a finite-dimensional vector space with ordered basis $B = (b_1, b_2, \dots, b_n)$. The map $T: V \rightarrow \mathbb{R}^n$ defined by $T(v) = v_B$, the coordinate vector of v relative to B , is an isomorphism.

PROOF Equation 1 in Section 3.3 shows that T preserves addition and scalar multiplication. Moreover, T is one-to-one, because the coordinate vector v_B of v uniquely determines v , and the range of T is all of $\mathbb{R}^n$. Therefore, T is an isomorphism. $\blacktriangle$

The isomorphism of V with $\mathbb{R}^n$, described in Theorem 3.9, is by no means unique. There is one such isomorphism for each choice of an ordered basis B of V .

Let V and V' be vector spaces of dimensions n and m , respectively. By Theorem 3.9, we can choose ordered bases B for V and B' for V' , and essentially convert V into $\mathbb{R}^n$ and V' into $\mathbb{R}^m$ by renaming vectors by their coordinate vectors relative to these bases. Then each linear transformation $T: V \rightarrow V'$ corresponds to a linear transformation $\bar{T}: \mathbb{R}^n \rightarrow \mathbb{R}^m$ in a natural way, and we can answer questions about T by studying $\bar{T}$ instead. But we can study $\bar{T}$ in turn by studying its standard matrix representation A , as we will

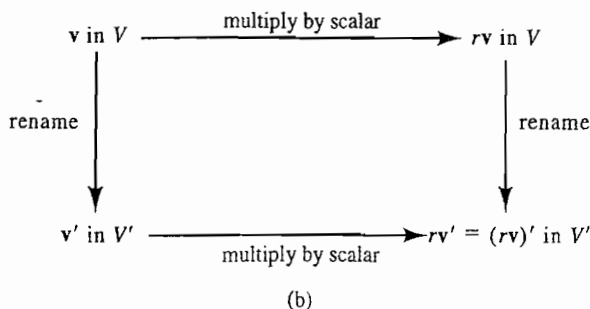
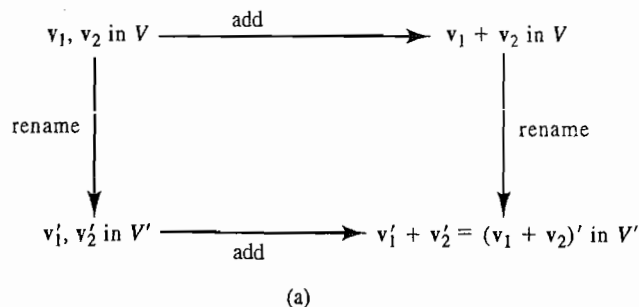


FIGURE 3.2

(a) The equal sign shows that the renaming preserves vector addition.

(b) The equal sign shows that the renaming preserves scalar multiplication.

illustrate shortly. This matrix changes if we change the ordered bases B or B' . A good deal of the remainder of our text is devoted to studying how to choose B and B' when $m = n$ so that the square matrix A has a simple form that illuminates the structure of the transformation T . This is the thrust of Chapters 5 and 7.

Matrix Representations of Transformations

In Section 2.3, we saw that every linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ can be computed using its standard matrix representation A , where A is the $m \times n$ matrix having as j th column vector $T(\mathbf{e}_j)$. Namely, for this matrix A and any column vector $\mathbf{x} \in \mathbb{R}^n$, we have $T(\mathbf{x}) = A\mathbf{x}$.

We just showed that finite-dimensional vector spaces V and V' , where $\dim(V) = n$ and $\dim(V') = m$, are isomorphic to $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively. We can rename vectors in V and V' by their coordinate vectors relative to ordered bases $B = (\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n)$ of V and $B' = (\mathbf{b}'_1, \mathbf{b}'_2, \dots, \mathbf{b}'_m)$ of V' , and work with the coordinates, doing computations in $\mathbb{R}^n$ and $\mathbb{R}^m$. In particular, a linear transformation $T: V \rightarrow V'$ and a choice of ordered bases B and B' then gives a linear transformation $\bar{T}: \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that for $\mathbf{v} \in V$ we have $\bar{T}(\mathbf{v}_B) = T(\mathbf{v})_{B'}$. This transformation $\bar{T}$ has an $m \times n$ standard matrix representation whose j th column vector is $\bar{T}(\mathbf{e}_j)$. Now $\mathbf{e}_j$ is the coordinate vector, relative to B , of the j th vector $\mathbf{b}_j$ in the ordered basis B , so $\bar{T}(\mathbf{e}_j) = \bar{T}((\mathbf{b}_j)_B) = T(\mathbf{b}_j)_{B'}$. Thus we see that the standard matrix representation of $\bar{T}$ is

$$A = \begin{bmatrix} | & | & & | \\ T(\mathbf{b}_1)_{B'} & T(\mathbf{b}_2)_{B'} & \cdots & T(\mathbf{b}_n)_{B'} \\ | & | & & | \end{bmatrix}. \quad (8)$$

We summarize in a theorem.

THEOREM 3.10 Matrix Representations of Linear Transformations

Let V and V' be finite-dimensional vector spaces and let $B = (\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n)$ and $B' = (\mathbf{b}'_1, \mathbf{b}'_2, \dots, \mathbf{b}'_m)$ be ordered bases for V and V' , respectively. Let $T: V \rightarrow V'$ be a linear transformation, and let $\bar{T}: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be the linear transformation such that for each $\mathbf{v} \in V$, we have $\bar{T}(\mathbf{v}_B) = T(\mathbf{v})_{B'}$. Then the standard matrix representation of $\bar{T}$ is the matrix A whose j th column vector is $T(\mathbf{b}_j)_{B'}$, and $T(\mathbf{v})_{B'} = A\mathbf{v}_B$ for all vectors $\mathbf{v} \in V$.

DEFINITION 3.11 Matrix Representation of T Relative to B, B'

The matrix A in Eq. (8) and described in Theorem 3.10 is the **matrix representation of T relative to B, B'** .

EXAMPLE 8 Let V be the subspace $\text{sp}(\sin x \cos x, \sin^2 x, \cos^2 x)$ of the vector space D of all differentiable functions mapping $\mathbb{R}$ into $\mathbb{R}$. Differentiation gives a linear transformation T of V into itself. Find the matrix representation A of T relative to B, B' where $B = B' = (\sin x \cos x, \sin^2 x, \cos^2 x)$. Use A to compute the derivative of

$$f(x) = 3 \sin x \cos x - 5 \sin^2 x + 7 \cos^2 x.$$

SOLUTION We find that

$$T(\sin x \cos x) = -\sin^2 x + \cos^2 x \quad \text{so} \quad T(\sin x \cos x)_{B'} = [0, -1, 1],$$

$$T(\sin^2 x) = 2 \sin x \cos x \quad \text{so} \quad T(\sin^2 x)_{B'} = [2, 0, 0], \quad \text{and}$$

$$T(\cos^2 x) = -2 \sin x \cos x \quad \text{so} \quad T(\cos^2 x)_{B'} = [-2, 0, 0].$$

Using Eq. (8), we have

$$A = \begin{bmatrix} 0 & 2 & -2 \\ -1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad \text{so} \quad A(f(x))_B = \begin{bmatrix} 0 & 2 & -2 \\ -1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ -5 \\ 7 \end{bmatrix} = \begin{bmatrix} -24 \\ -3 \\ 3 \end{bmatrix}.$$

Thus $f'(x) = -24 \sin x \cos x - 3 \sin^2 x + 3 \cos^2 x$. ■

EXAMPLE 9 Letting P_n be the vector space of polynomials of degree at most n , we note that $T: P_2 \rightarrow P_3$ defined by $T(p(x)) = (x+1)p(x-2)$ is a linear transformation. Find the matrix representation A of T relative to the ordered bases $B = (x^2, x, 1)$ and $B' = (x^3, x^2, x, 1)$ for P_2 and P_3 , respectively. Use A to compute $T(p(x))$ for $p(x) = 5x^2 - 7x + 18$.

SOLUTION We compute

$$T(x^2) = (x+1)(x-2)^2 = (x+1)(x^2 - 4x + 4) = x^3 - 3x^2 + 4,$$

$$T(x) = (x+1)(x-2) = x^2 - x - 2, \quad \text{and} \quad T(1) = (x+1)1 = x + 1.$$

Thus $T(x^2)_{B'} = [1, -3, 0, 4]$, $T(x)_{B'} = [0, 1, -1, -2]$, and $T(1)_{B'} = [0, 0, 1, 1]$. Consequently,

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & -1 & 1 \\ 4 & -2 & 1 \end{bmatrix} \quad \text{so} \quad A(p(x))_B = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & -1 & 1 \\ 4 & -2 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ -7 \\ 18 \end{bmatrix} = \begin{bmatrix} 5 \\ -22 \\ 25 \\ 52 \end{bmatrix}.$$

Thus $T(p(x)) = 5x^3 - 22x^2 + 25x + 52$. ■

Let V and V' be n -dimensional vector spaces with ordered bases B and B' , respectively. Theorem 3.8 tells us that a linear transformation $T: V \rightarrow V'$ is invertible if and only if it is one-to-one and onto V' . Under these circumstances, we might expect the following to be true:

Matrix Representation of T^{-1}

The matrix representation of T^{-1} relative to B', B is the inverse of the matrix representation of T relative to B, B' .

This is indeed the case in view of Theorem 3.10 and the fact that linear transformations of $\mathbb{R}^n$ have this property. (See the box on page 151.) Exercise 20 gives an illustration of this.

It is important to note that the matrix representation of the linear transformation $T: V \rightarrow V'$ depends on the particular bases B for V and B' for V' . We really should use some notation such as $A_{B, B'}$ or $R_{B, B'}$ to denote this dependency. Such notations appear cumbersome and complicated, so we avoid them for now. We will use such a notation in Chapter 7, where we will consider this dependency in more detail.

SUMMARY

Let V and V' be vector spaces, and let T be a function mapping V into V' .

1. The function T is a linear transformation if it preserves addition and scalar multiplication—that is, if

$$T(\mathbf{v}_1 + \mathbf{v}_2) = T(\mathbf{v}_1) + T(\mathbf{v}_2)$$

and

$$T(r\mathbf{v}_1) = rT(\mathbf{v}_1)$$

for all vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ in V and for scalars r in $\mathbb{R}$.

2. If T is a linear transformation, then $T(\mathbf{0}) = \mathbf{0}'$ and also $T(\mathbf{v}_1 - \mathbf{v}_2) = T(\mathbf{v}_1) - T(\mathbf{v}_2)$.
3. A function $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation if and only if T has the form $T(\mathbf{x}) = A\mathbf{x}$ for some $m \times n$ matrix A .
4. The matrix A in summary item 3 is the standard matrix representation of the transformation T .
5. A linear transformation $T: V \rightarrow V'$ is invertible if and only if it is one-to-one and onto V' . Such transformations are isomorphisms.
6. Every nonzero finite-dimensional real vector space V is isomorphic to $\mathbb{R}^n$, where $n = \dim(V)$.

Now let $T: V \rightarrow V'$ be a linear transformation.

7. If W is a subspace of V , then $T[W]$ is a subspace of V' . If W' is a subspace of V' , then $T^{-1}[W']$ is a subspace of V .

8. The range of T is the subspace $\{T(\mathbf{v}) \mid \mathbf{v} \in V\}$ of V' , and the kernel $\ker(T)$ is the subspace $\{\mathbf{v} \in V \mid T(\mathbf{v}) = \mathbf{0}\}$ of V .
9. T is one-to-one if and only if $\ker(T) = \{\mathbf{0}\}$. If T is one-to-one and has as its range all of V' , then $T^{-1}: V' \rightarrow V$ is well-defined and is a linear transformation. In this case, both T and T^{-1} are isomorphisms.
10. If $T: V \rightarrow V'$ is a linear transformation and $T(\mathbf{p}) = \mathbf{b}$, then the solution set of $T(\mathbf{x}) = \mathbf{b}$ is $\{\mathbf{p} + \mathbf{h} \mid \mathbf{h} \in \ker(T)\}$.
11. Let $B = (\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n)$ and $B' = (\mathbf{b}'_1, \mathbf{b}'_2, \dots, \mathbf{b}'_m)$ be ordered bases for V and V' , respectively. The matrix representation of T relative to B, B' is the $m \times n$ matrix A having $T(\mathbf{b}_j)_{B'}$ as its j th column vector. We have $T(\mathbf{v})_{B'} = A(\mathbf{v}_B)$ for all $\mathbf{v} \in V$.

EXERCISES

In Exercises 1–5, let F be the vector space of all functions mapping $\mathbb{R}$ into $\mathbb{R}$. Determine whether the given function T is a linear transformation. If it is a linear transformation, describe the kernel of T and determine whether the transformation is invertible.

1. $T: F \rightarrow \mathbb{R}$ defined by $T(f) = f(-4)$
2. $T: F \rightarrow \mathbb{R}$ defined by $T(f) = f(5)^2$
3. $T: F \rightarrow F$ defined by $T(f) = f + f$
4. $T: F \rightarrow F$ defined by $T(f) = f + 3$, where 3 is the constant function with value 3 for all $x \in \mathbb{R}$.
5. $T: F \rightarrow F$ defined by $T(f) = -f$
6. Let $C_{0,2}$ be the space of continuous functions mapping the interval $0 \leq x \leq 2$ into $\mathbb{R}$. Let $T: C_{0,2} \rightarrow \mathbb{R}$ be defined by $T(f) = \int_0^2 f(x) dx$. See Example 3. If possible, give three different functions in $\ker(T)$.
7. Let C be the space of all continuous functions mapping $\mathbb{R}$ into $\mathbb{R}$, and let $T: C \rightarrow C$ be defined by $T(f) = \int_1^x f(t) dt$. See Example 4. If possible, give three different functions in $\ker(T)$.
8. Let F be the vector space of all functions mapping $\mathbb{R}$ into $\mathbb{R}$, and let $T: F \rightarrow F$ be a linear transformation such that $T(e^{2x}) = x^2$, $T(e^{3x}) = \sin x$, and $T(1) = \cos 5x$. Find the following, if it is determined by this data.
 - a. $T(e^{5x})$
 - c. $T(3e^{4x})$
 - b. $T(3 + 5e^{3x})$
 - d. $T\left(\frac{e^{4x} + 2e^{3x}}{e^{2x}}\right)$

9. Note that one solution of the differential equation $y'' - 4y = \sin x$ is $y = -\frac{1}{5} \sin x$. Use summary item 10 and Illustration 1 to describe all solutions of this equation.

Let D_n be the vector space of functions mapping $\mathbb{R}$ into $\mathbb{R}$ that have derivatives of all orders. It can be shown that the kernel of a linear transformation $T: D_n \rightarrow D_n$ of the form $T(f) = a_n f^{(n)} + \dots + a_1 f' + a_0 f$ where $a_n \neq 0$ is an n -dimensional subspace of D_n .

In Exercises 10–15, use the preceding information, summary item 10, and your knowledge of calculus to find all solutions in D_n of the given differential equation. See Illustration 1 in the text.

10. $y' = \sin 2x$
11. $y'' = -\cos x$
12. $y' - y = x$
13. $y'' + 4y = x^2$
14. $y'' + y' = 3e^x$
15. $y^{(3)} - 2y'' = x$
16. Let V and V' be vector spaces having ordered bases $B = (\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3)$ and $B' = (\mathbf{b}'_1, \mathbf{b}'_2, \mathbf{b}'_3, \mathbf{b}'_4)$, respectively. Let $T: V \rightarrow V'$ be a linear transformation such that

$$\begin{aligned} T(\mathbf{b}_1) &= 3\mathbf{b}'_1 + \mathbf{b}'_2 + 4\mathbf{b}'_3 - \mathbf{b}'_4 \\ T(\mathbf{b}_2) &= \mathbf{b}'_1 + 2\mathbf{b}'_2 - \mathbf{b}'_3 + 2\mathbf{b}'_4 \\ T(\mathbf{b}_3) &= -2\mathbf{b}'_1 - \mathbf{b}'_2 + 2\mathbf{b}'_3 \end{aligned}$$

Find the matrix representation A of T relative to B, B' .

In Exercises 17–19, let V and V' be vector spaces with ordered bases $B = (b_1, b_2, b_3)$ and $B' = (b'_1, b'_2, b'_3, b'_4)$, respectively, and let $T: V \rightarrow V'$ be the linear transformation having the given matrix A as matrix representation relative to B, B' . Find $T(v)$ for the given vector v .

$$17. A = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 2 & 0 \\ 0 & 6 & 1 \\ 2 & 1 & 3 \end{bmatrix}, v = b_1 + b_2 + b_3$$

$$18. A \text{ as in Exercise 17, } v = 3b_3 - b_1$$

$$19. A = \begin{bmatrix} 0 & 4 & -1 \\ 1 & 1 & 2 \\ 2 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, v = 6b_1 - 4b_2 + b_3$$

In Exercises 20–33, we consider A to be the matrix representation of the indicated linear transformation T (you may assume it is linear) relative to the indicated ordered bases B and B' . Starting with Exercise 21, we let D denote differentiating once, D^2 differentiating twice, and so on.

20. Let V and V' be vector spaces with ordered bases $B = (b_1, b_2, b_3)$ and $B' = (b'_1, b'_2, b'_3)$, respectively. Let $T: V \rightarrow V'$ be the linear transformation such that

$$T(b_1) = b'_1 + 2b'_2 - 3b'_3$$

$$T(b_2) = 3b'_1 + 5b'_2 + 2b'_3$$

$$T(b_3) = -2b'_1 - 3b'_2 - 4b'_3.$$

- Find the matrix A .
 - Use A to find $T(v)$, if $v_B = [2, -5, 1]$.
 - Show that T is invertible, and find the matrix representation of T^{-1} relative to B', B .
 - Find $T^{-1}(v')$ if $v'_B = [-1, 1, 3]$.
 - Express $T^{-1}(b'_1)$, $T^{-1}(b'_2)$, and $T^{-1}(b'_3)$ as linear combinations of the vectors in B .
21. Let $T: P_3 \rightarrow P_3$ be defined by $T(p(x)) = D(p(x))$, the derivative of $p(x)$. Let the ordered bases for P_3 be $B = B' = (x^3, x^2, x, 1)$.
- Find the matrix A .
 - Use A to find the derivative of $4x^3 - 5x^2 + 10x - 13$.
 - Noting that $T \circ T = D^2$, find the second derivative of $-5x^3 + 8x^2 - 3x + 4$ by multiplying a column vector by an appropriate matrix.
22. Let $T: P_3 \rightarrow P_3$ be defined by $T(p(x)) = xD(p(x))$ and let the ordered bases B and B' be as in Exercise 21.
- Find the matrix representation A relative to B, B' .
 - Working with the matrix A and coordinate vectors, find all solutions $p(x)$ of $T(p(x)) = x^3 - 3x^2 + 4x$.
 - The transformation T can be decomposed into $T = T_2 \circ T_1$, where $T_1: P_3 \rightarrow P_2$ is defined by $T_1(p(x)) = D(p(x))$ and $T_2: P_2 \rightarrow P_3$ is defined by $T_2(p(x)) = xp(x)$. Find the matrix representations of T_1 and T_2 using the ordered bases B of P_3 and $B'' = (x^2, x, 1)$ of P_2 . Now multiply these matrix representations for T_1 and T_2 to obtain a 4×4 matrix, and compare with the matrix A . What do you notice?
 - Multiply the two matrices found in part (c) to obtain a 3×3 matrix. Let $T_3: P_2 \rightarrow P_2$ be the linear transformation having this matrix as matrix representation relative to B'', B'' for the ordered basis B'' of part (c). Find $T_3(a_2x^2 + a_1x + a_0)$. How is T_3 related to T_1 and T_2 ?
23. Let V be the subspace $\text{sp}(x^2e^x, xe^x, e^x)$ of the vector space of all differentiable functions mapping $\mathbb{R}$ into $\mathbb{R}$. Let $T: V \rightarrow V$ be the linear transformation of V into itself given by taking second derivatives, so $T = D^2$, and let $B = B' = (x^2e^x, xe^x, e^x)$. Find the matrix A by
- following the procedure in summary item 11, and
 - finding and then squaring the matrix A_1 that represents the transformation D corresponding to taking first derivatives.
24. Let $T: P_3 \rightarrow P_2$ be defined by $T(p(x)) = p'(2x + 1)$, where $p'(x) = D(p(x))$, and let $B = (x^3, x^2, x, 1)$ and $B' = (x^2, x, 1)$.
- Find the matrix A .
 - Use A to compute $T(4x^3 - 5x^2 + 4x - 7)$.
25. Let $V = \text{sp}(\sin^2x, \cos^2x)$ and let $T: V \rightarrow V$ be defined by taking second derivatives.

Taking $B = B' = (\sin^2 x, \cos^2 x)$, find A in two ways.

- Compute A as described in summary item 11.
 - Find the space W spanned by the first derivatives of the vectors in B , choose an ordered basis for W , and compute A as a product of the two matrices representing the differentiation map from V into W followed by the differentiation map from W into V .
26. Let $T: P_3 \rightarrow P_3$ be the linear transformation defined by $T(p(x)) = D^2(p(x)) - 4D(p(x)) + p(x)$. Find the matrix representation A of T , where $B = (x, 1 + x, x + x^2, x^3)$.
27. Let $W = \text{sp}(e^{2x}, e^{4x}, e^{8x})$ be the subspace of the vector space of all real-valued functions with domain $\mathbb{R}$, and let $B = (e^{2x}, e^{4x}, e^{8x})$. Find the matrix representation A relative to B , B of the linear transformation $T: W \rightarrow W$ defined by $T(f) = D^2(f) + 2D(f) + f$.
28. For W and B in Exercise 27, find the matrix representation A of the linear transformation $T: W \rightarrow W$ defined by $T(f) = \int_{-\infty}^x f(t) dt$.
29. For W and B in Exercise 27, find $T(ae^{2x} + be^{4x} + ce^{8x})$ for the linear transformation T whose matrix representation relative to B, B is

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}.$$

30. Repeat Exercise 29, given that

$$A = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & -3 \end{bmatrix}.$$

31. Let W be the subspace $\text{sp}(\sin 2x, \cos 2x)$ of the vector space of all real-valued functions with domain $\mathbb{R}$, and let $B = (\sin 2x, \cos 2x)$. Find the matrix representation A relative to B, B for the linear transformation $T: W \rightarrow W$ defined by $T(f) = D^2(f) + 2D(f) + f$.
32. For W and B in Exercise 31, find $T(a \sin 2x + b \cos 2x)$ for $T: W \rightarrow W$ whose matrix representation is

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

33. For W and B in Exercise 31, find $T(a \sin 2x + b \cos 2x)$ for $T: W \rightarrow W$ whose matrix representation is

$$A = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}.$$

34. Let V and V' be vector spaces. Mark each of the following True or False.
- A linear transformation of vector spaces preserves the vector-space operations.
 - Every function mapping V into V' relates the algebraic structure of V to that of V' .
 - A linear transformation $T: V \rightarrow V'$ carries the zero vector of V into the zero vector of V' .
 - A linear transformation $T: V \rightarrow V'$ carries a pair $v, -v$ in V into a pair $v', -v'$ in V' .
 - For every vector b' in V' , the function $T_{b'}: V \rightarrow V'$ defined by $T_{b'}(v) = b'$ for all v in V is a linear transformation.
 - The function $T_0: V \rightarrow V'$ defined by $T_0(v) = 0'$, the zero vector of V' , for all v in V is a linear transformation.
 - The vector space P_{10} of polynomials of degree ≤ 10 is isomorphic to $\mathbb{R}^{10}$.
 - There is exactly one isomorphism $T: P_{10} \rightarrow \mathbb{R}^{11}$.
 - Let V and V' be vector spaces of dimensions n and m , respectively. A linear transformation $T: V \rightarrow V'$ is invertible if and only if $m = n$.
 - If T in part (i) is an invertible transformation, then $m = n$.
35. Prove that the two conditions in Definition 3.9 for a linear transformation are equivalent to the single condition in Eq. (3).
36. Let V, V' , and V'' be vector spaces, and let $T: V \rightarrow V'$ and $T': V' \rightarrow V''$ be linear transformations. Prove that the composite function $(T' \circ T): V \rightarrow V''$ defined by $(T' \circ T)(v) = T'(T(v))$ for each v in V is again a linear transformation.
37. Prove that, if T and T' are invertible linear transformations of vector spaces such that $T' \circ T$ is defined, $T' \circ T$ is also invertible.
38. State conditions for an $m \times n$ matrix A that are equivalent to the condition that the linear transformation $T(x) = Ax$ for x in $\mathbb{R}^n$ is an isomorphism.